

Assembly Manual

CSAimBot

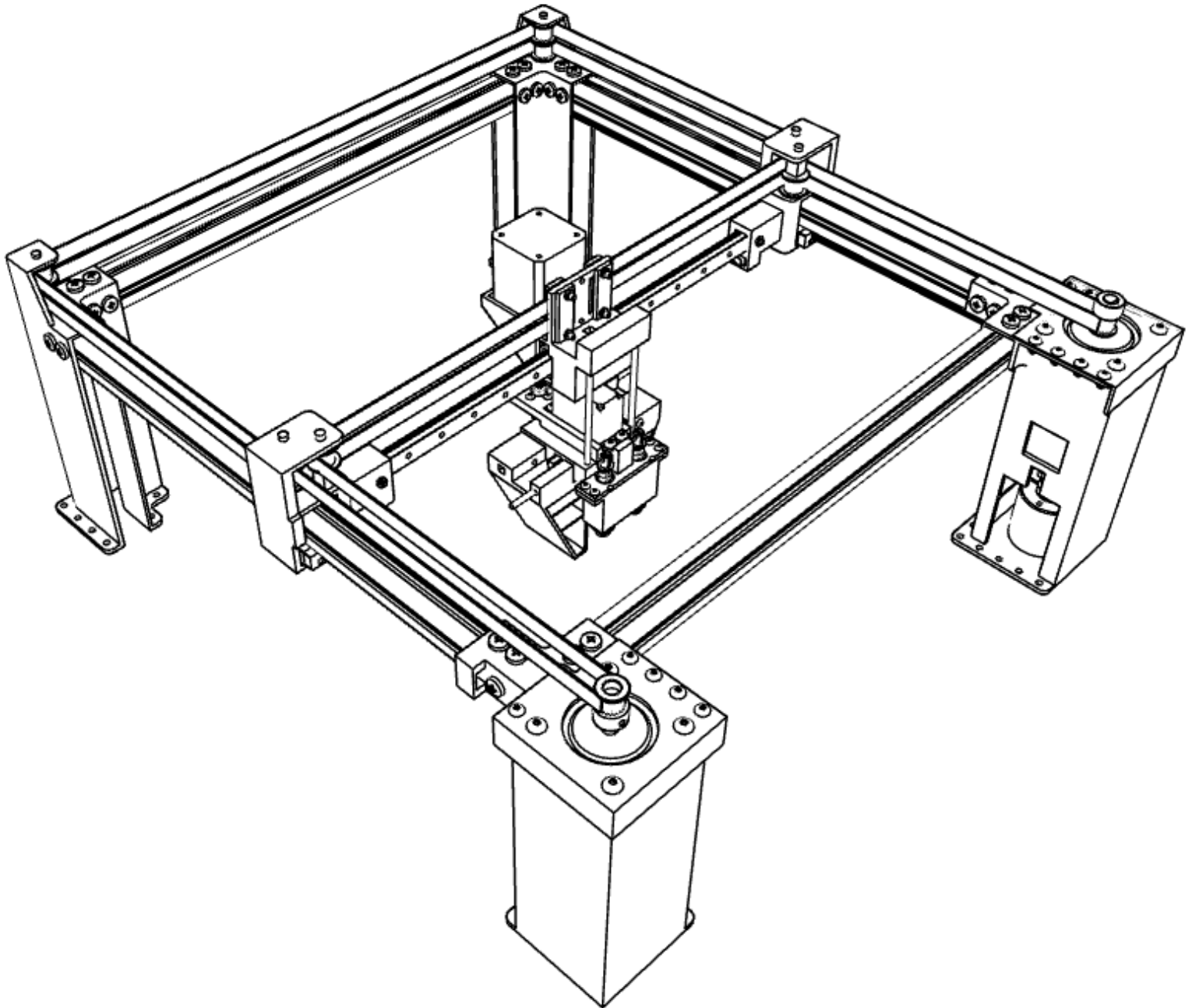


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Symbols used in the manual:



Apply glue (e.g. cyanoacrylate)

2x

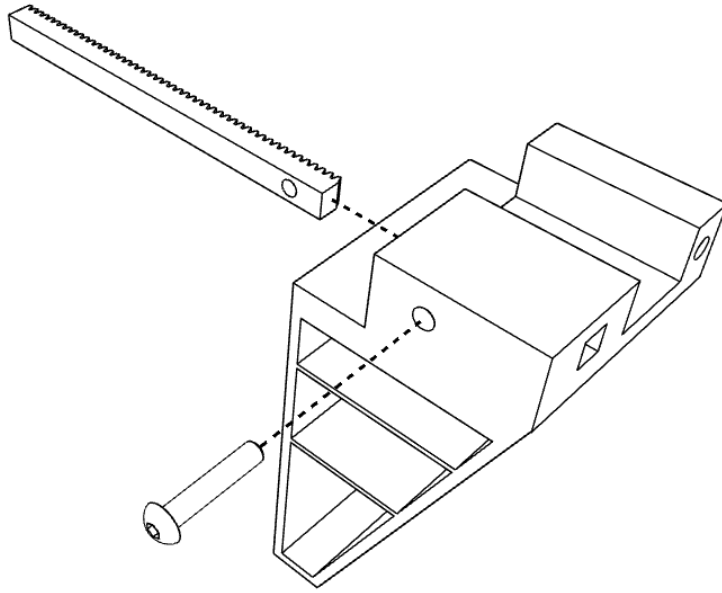
Repeat the action twice

1. Effector Module

Required Components:

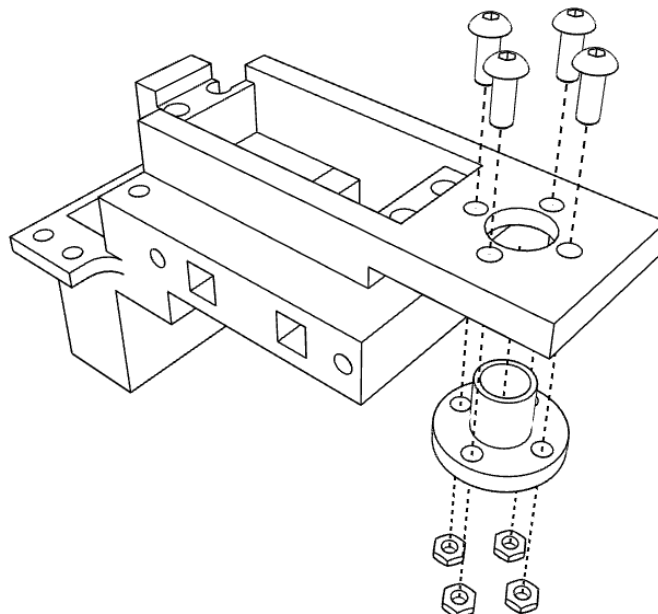
- Ø3 × 130 mm Linear Shaft x2
- Gear Rack x2
- Gripper Jaws x2
- TR8×2 Lead Screw Nut x1
- TR8×2 100 mm Lead Screw x1
- Jaw Mount Body x1
- Solenoid Cover x1
- Solenoid x2
- Solenoid Tips x2
- Servo Gear x1
- Servo Motor x1
- M3×10 mm Screw x12
- M3×16 mm Screw x2
- M3 Hex Nut x12

1

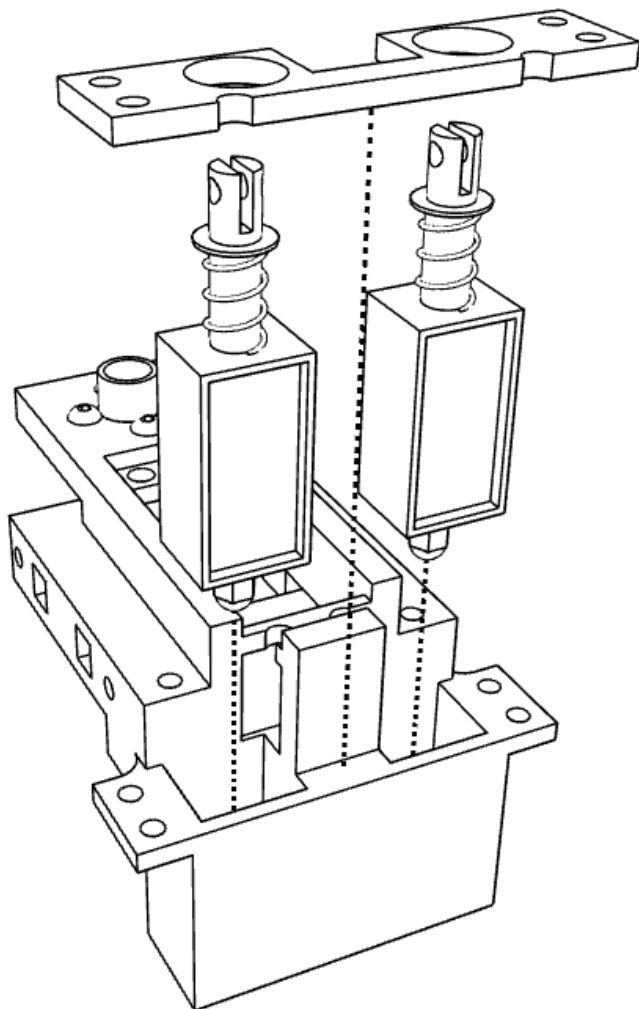


2x

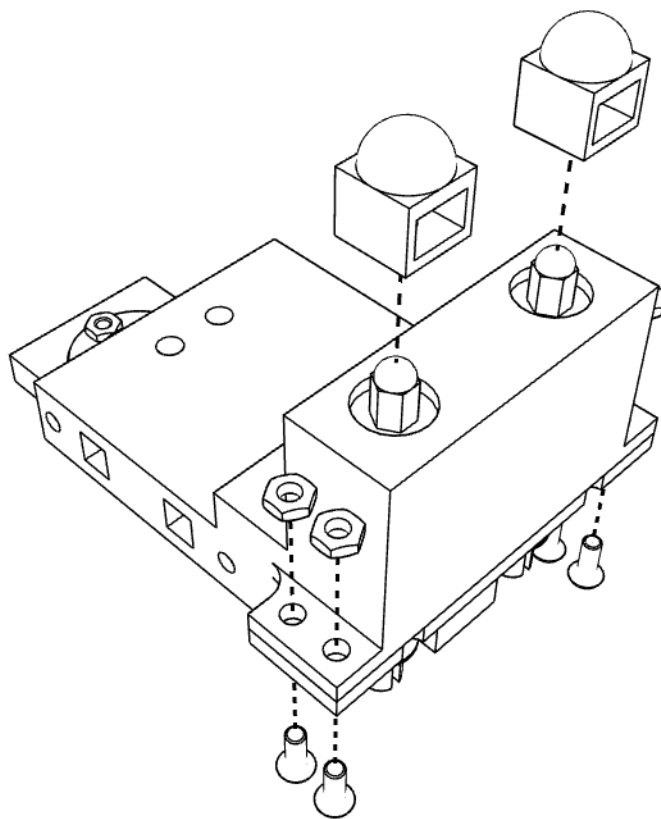
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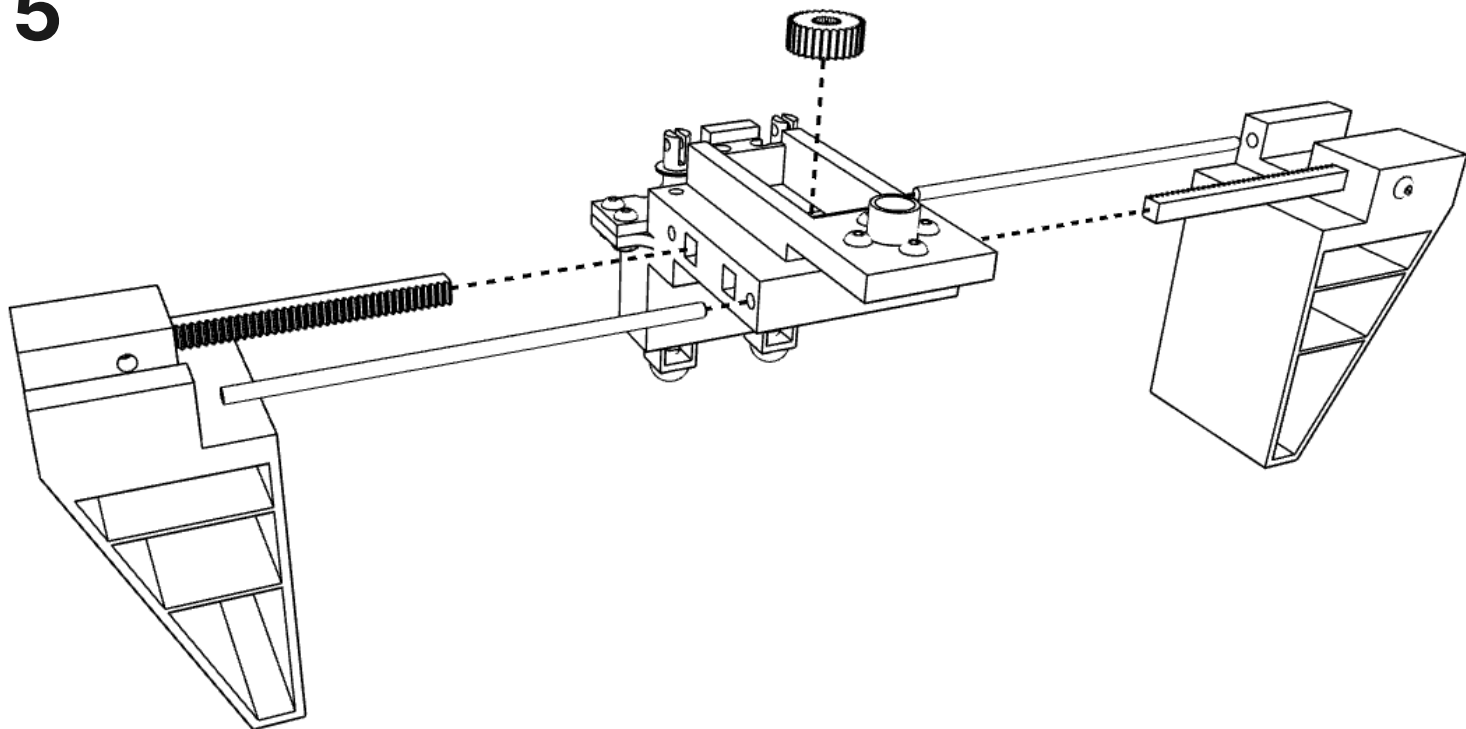
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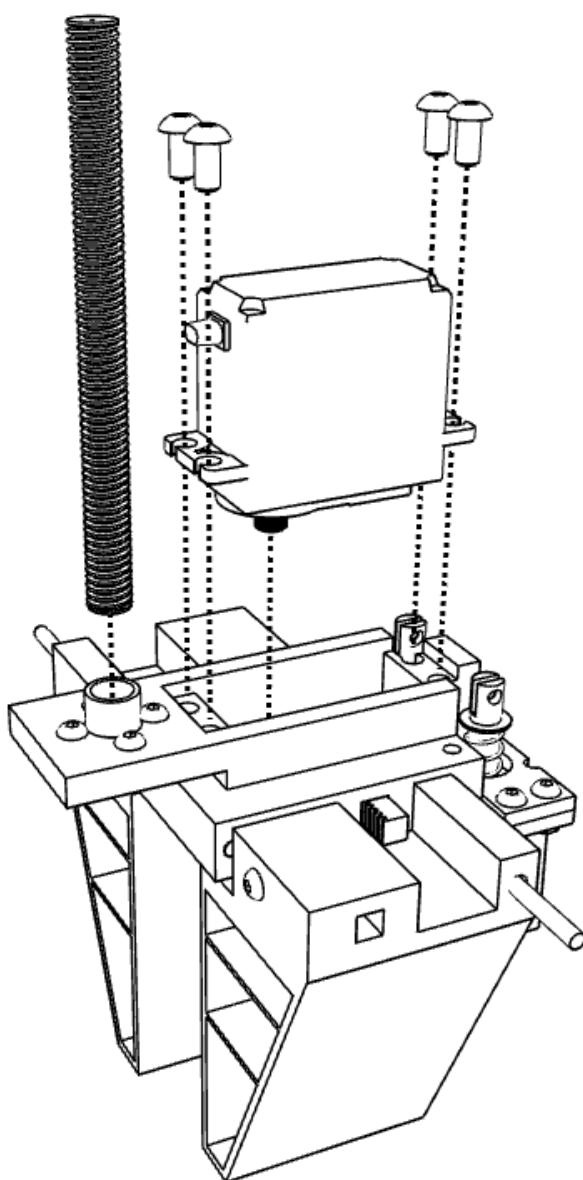
4



5



6

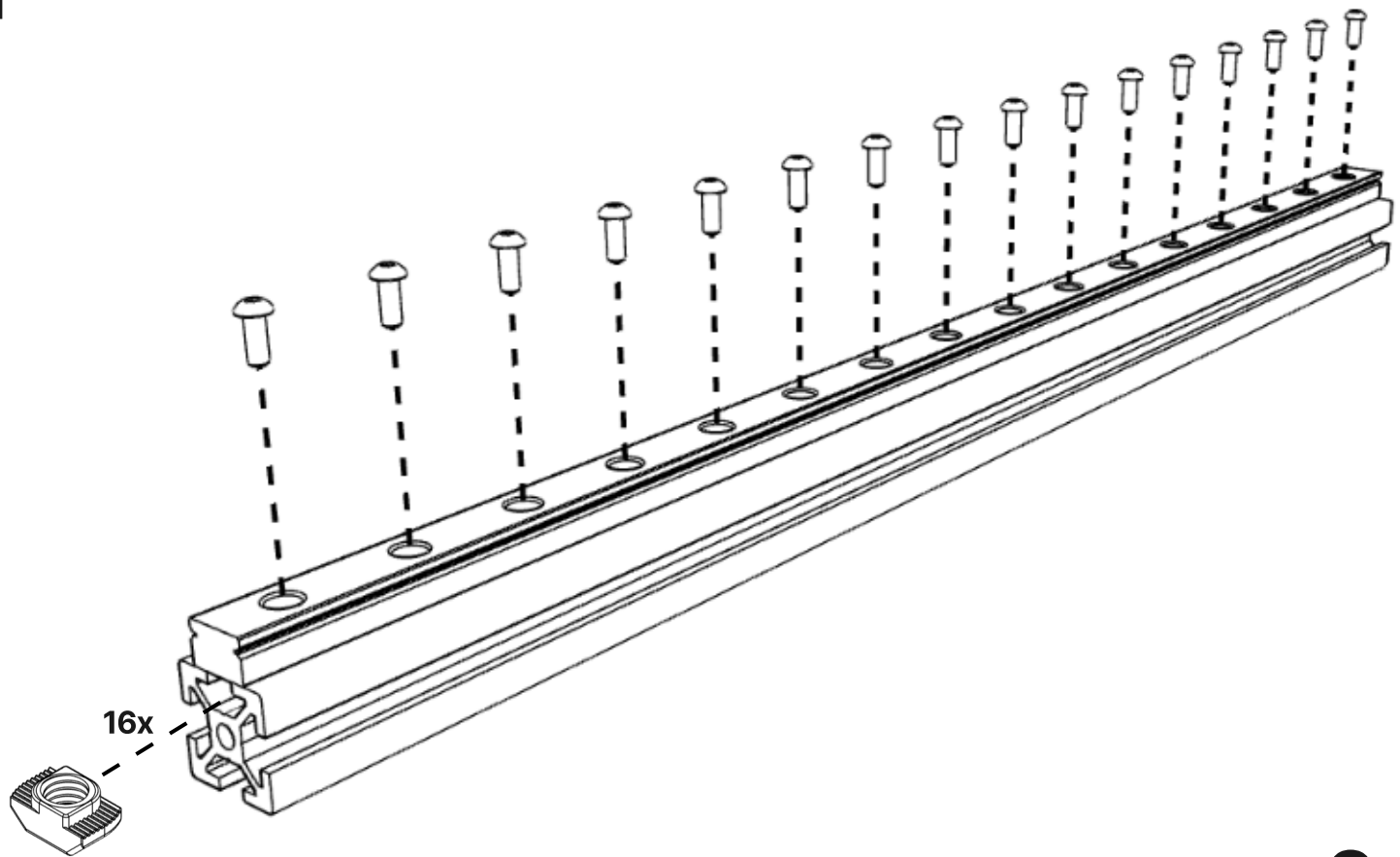


2. Linear Guide Module

Required Components:

- 400 mm Aluminum Structural Profile x2
- MGN12 400 mm Linear Guide Rail x2
- M3×10 mm Screw x32
- M3 T-Nut x32

1



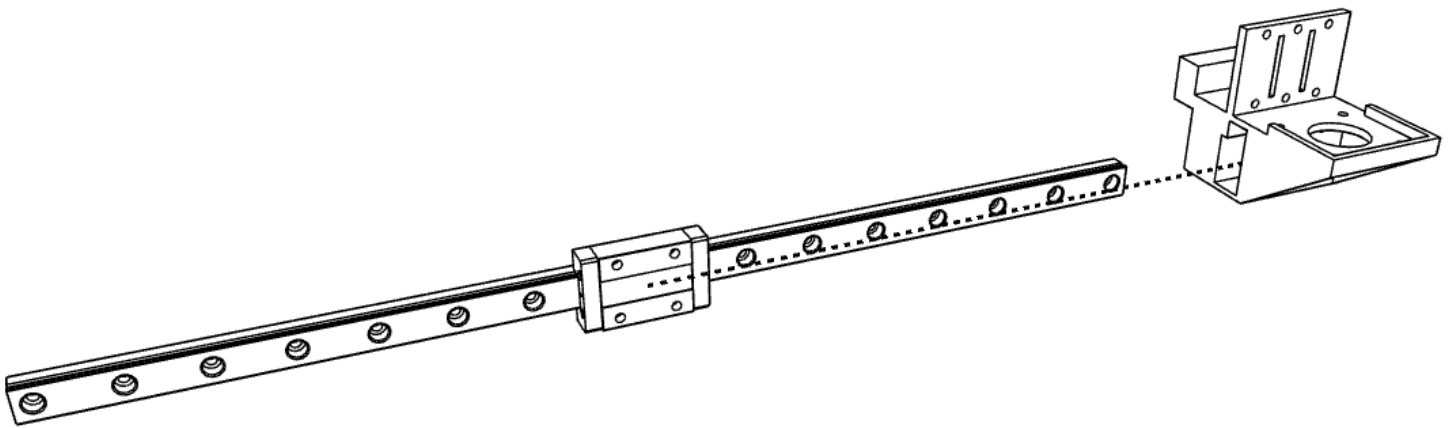
2x

3. Gripper Mount Module

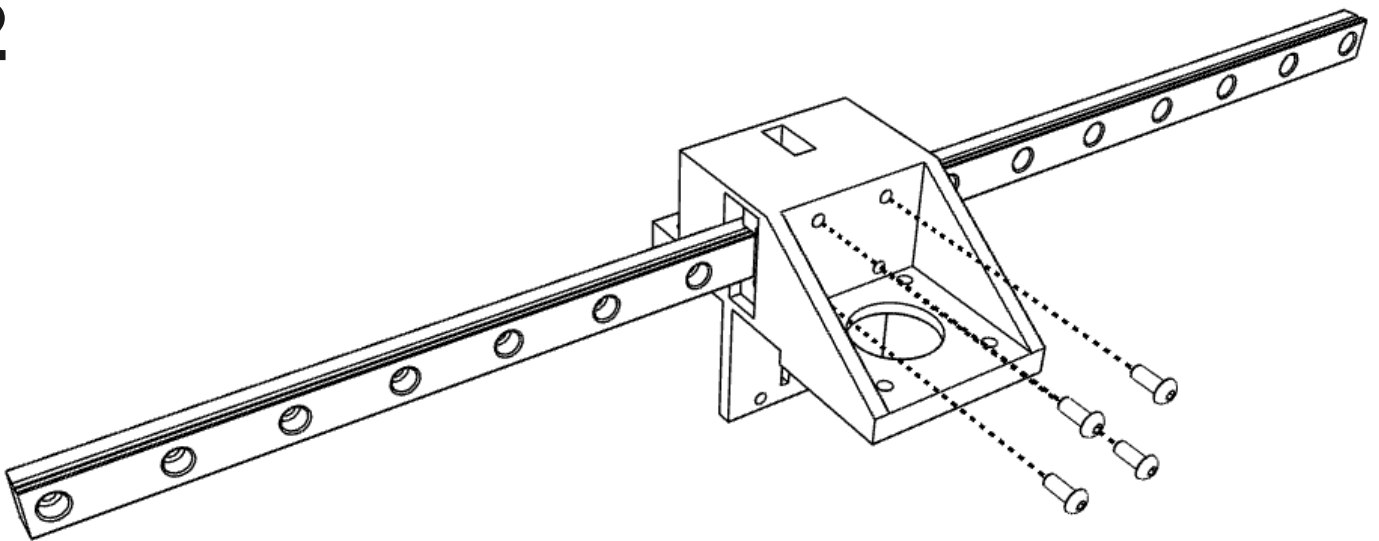
Required Components:

- Ø3×120 mm Linear Shaft x2
- Gripper Carriage Body x1
- MGN12 386 mm Linear Guide Rail x1
- Limit Switch x1
- M3×10 mm Screw x4

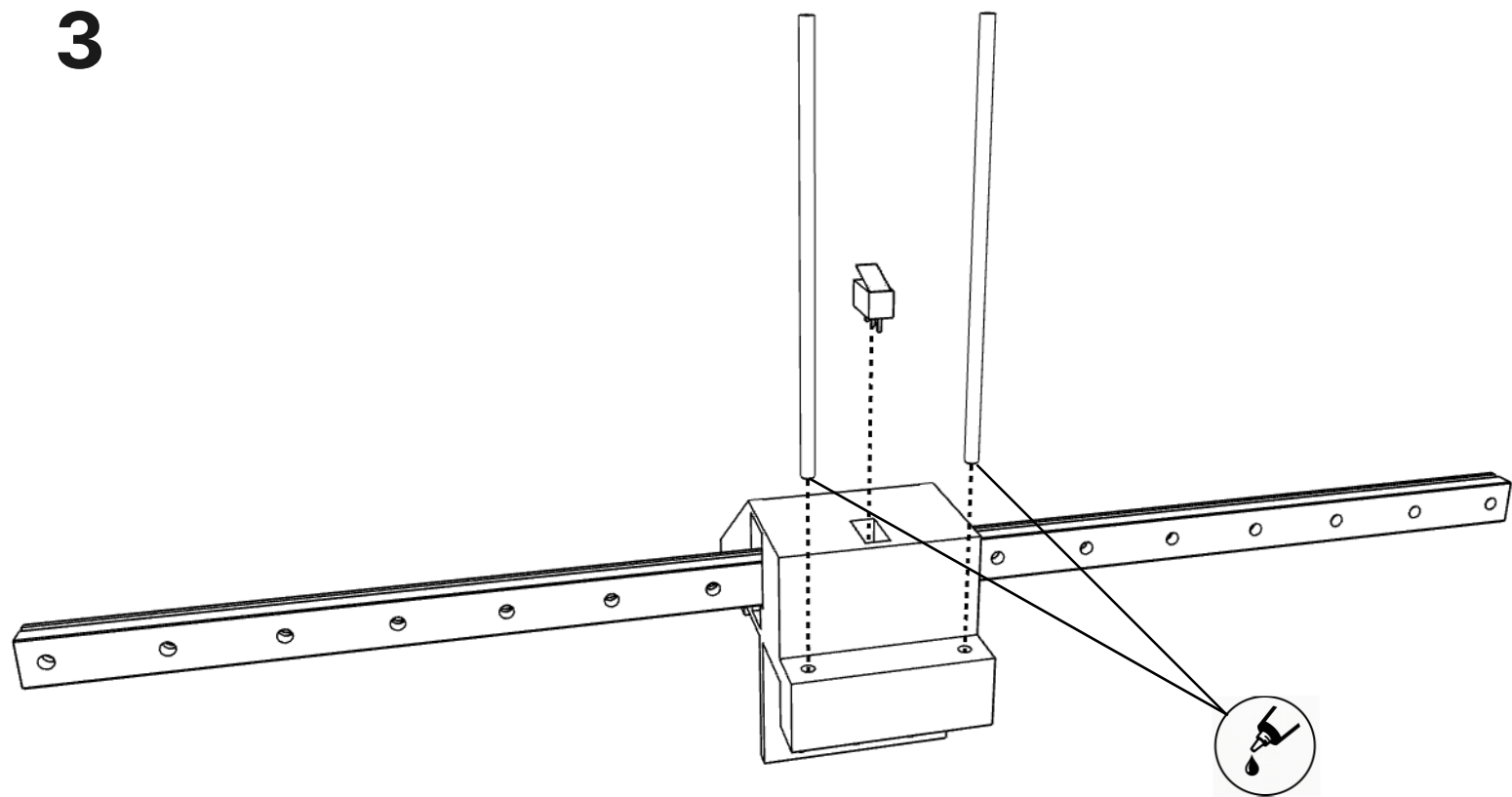
1



2



3

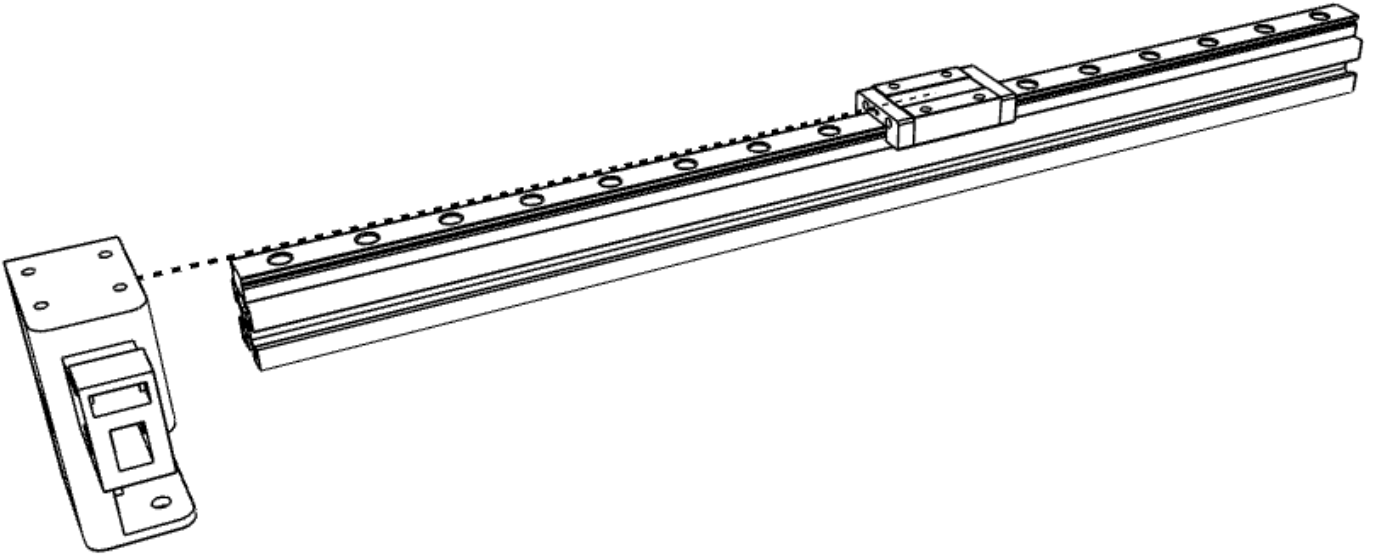


4. Linear Carriage Left Side

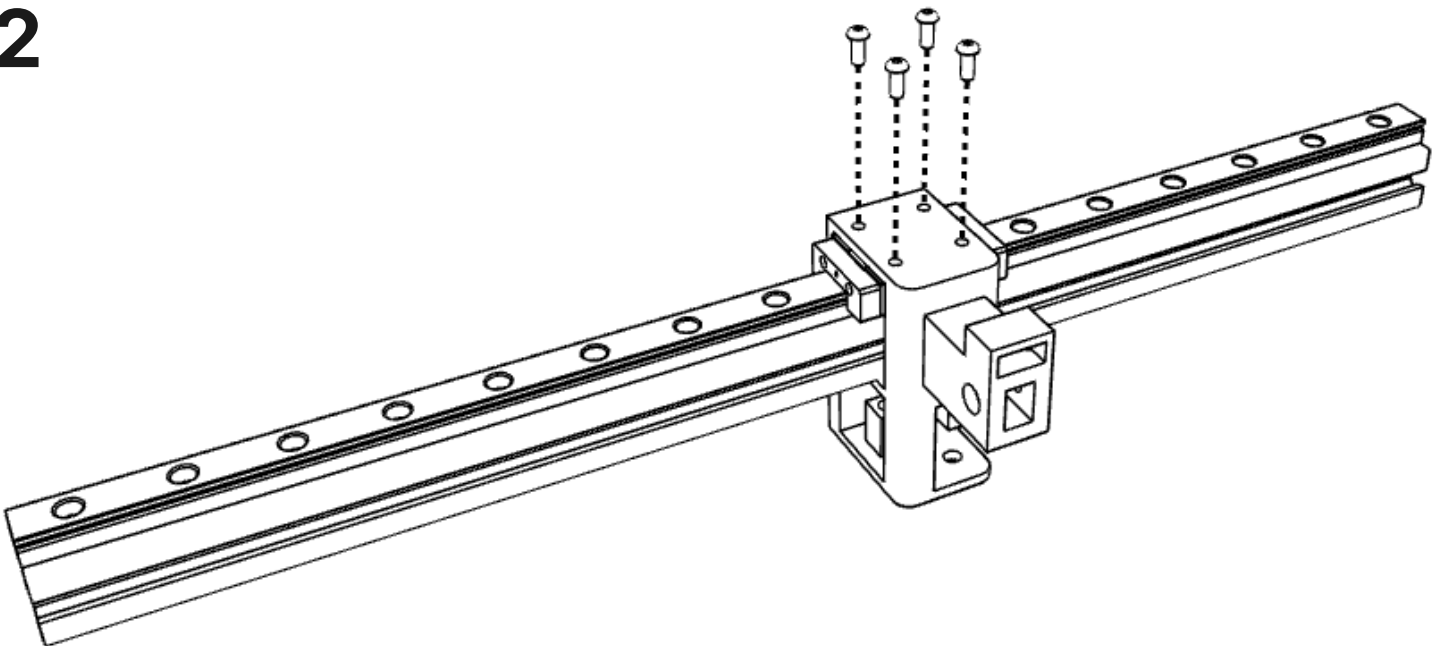
Required Components:

- Assembled Module from Step 2
- Linear Carriage Body Left x1
- Ø5×10 mm Linear Shaft x2
- GT2 Smooth Idler Pulley x1
- GT2 Toothed Idler Pulley x1
- M3×10 mm Screw x4

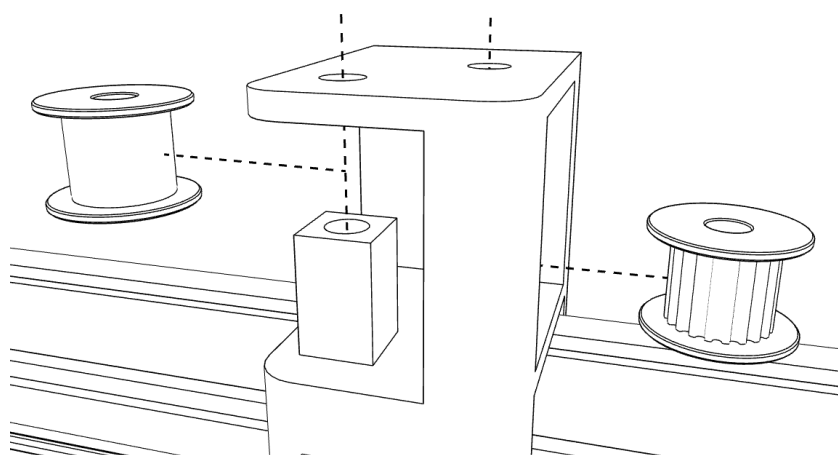
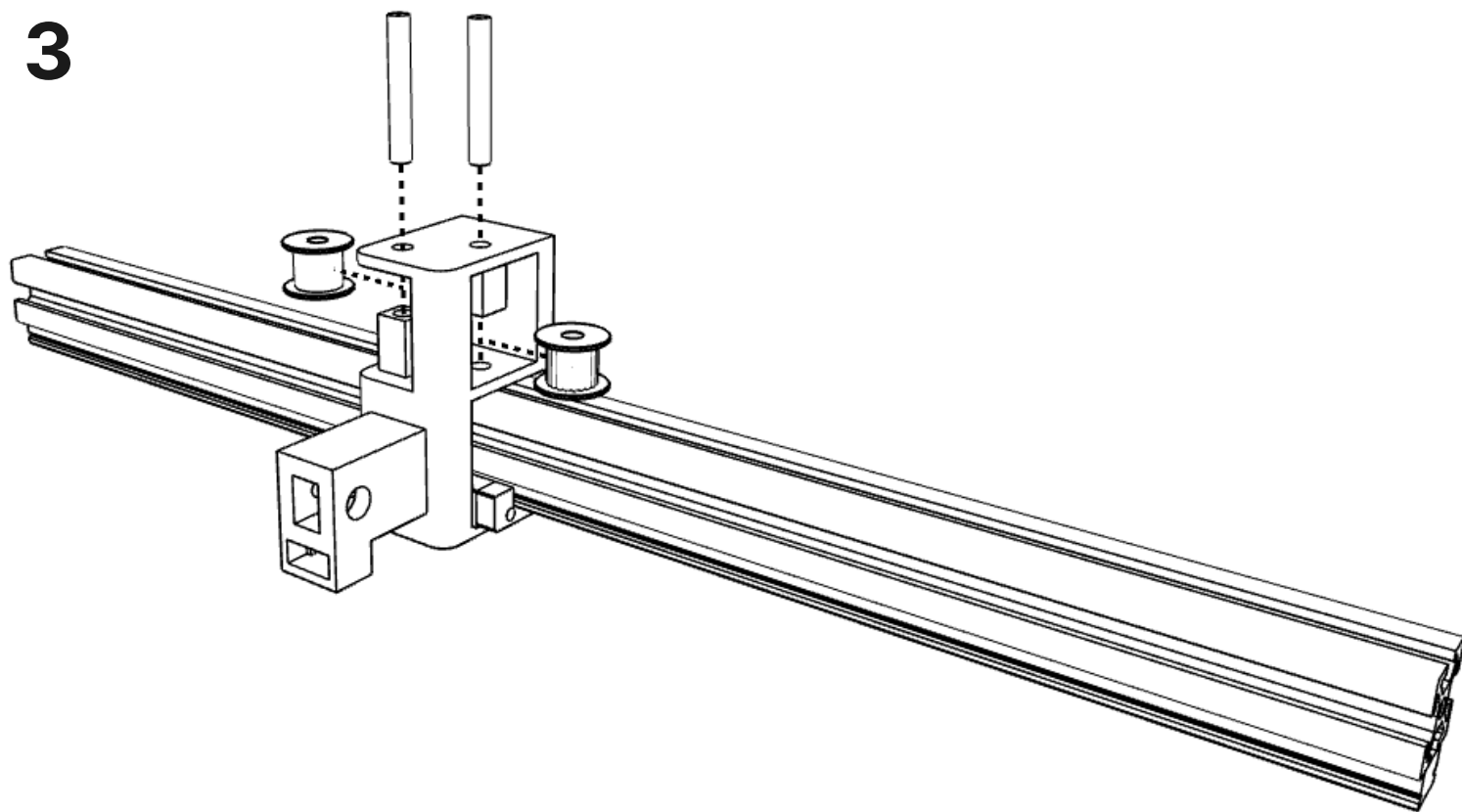
1



2



3

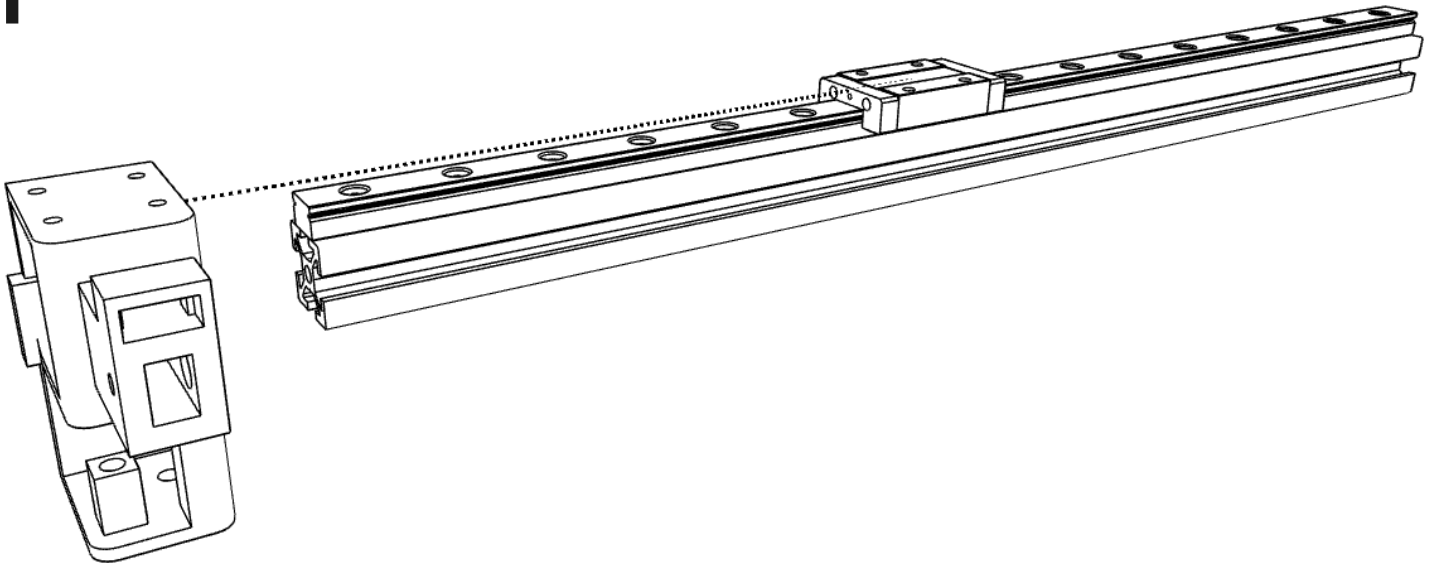


5. Linear Carriage Right Side

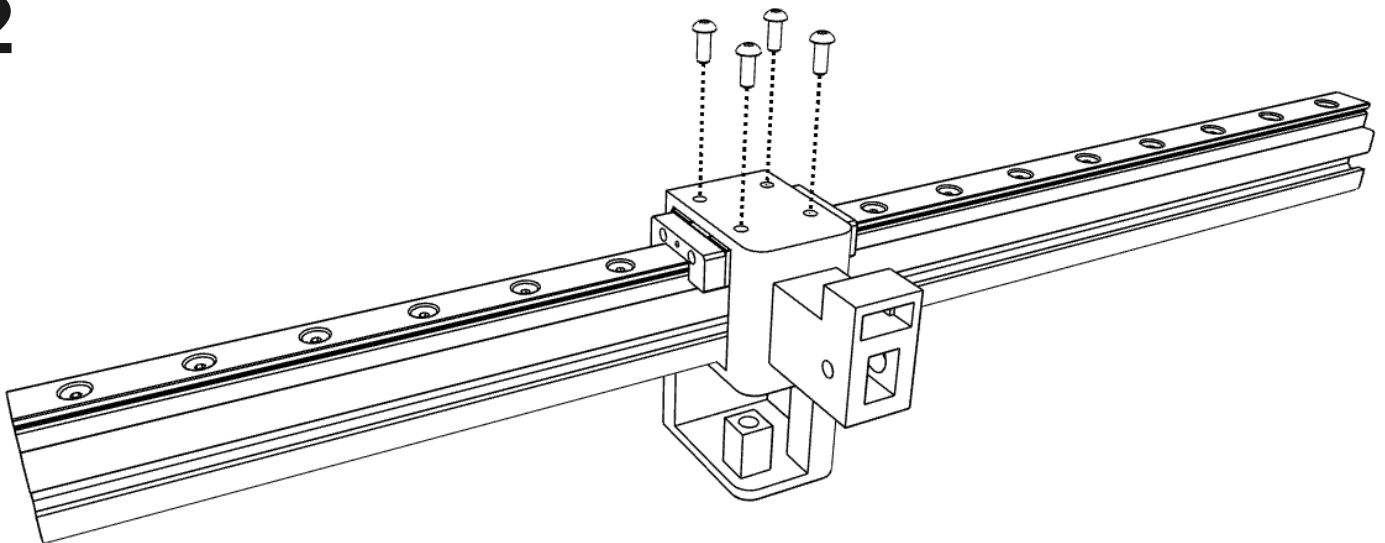
Required Components:

- Assembled Module from Step 2
- Linear Carriage Body Right x1
- Ø5×10 mm Linear Shaft x2
- GT2 Smooth Idler Pulley x1
- GT2 Toothed Idler Pulley x1
- Limit Switch x1
- M3×10 mm Screw x4

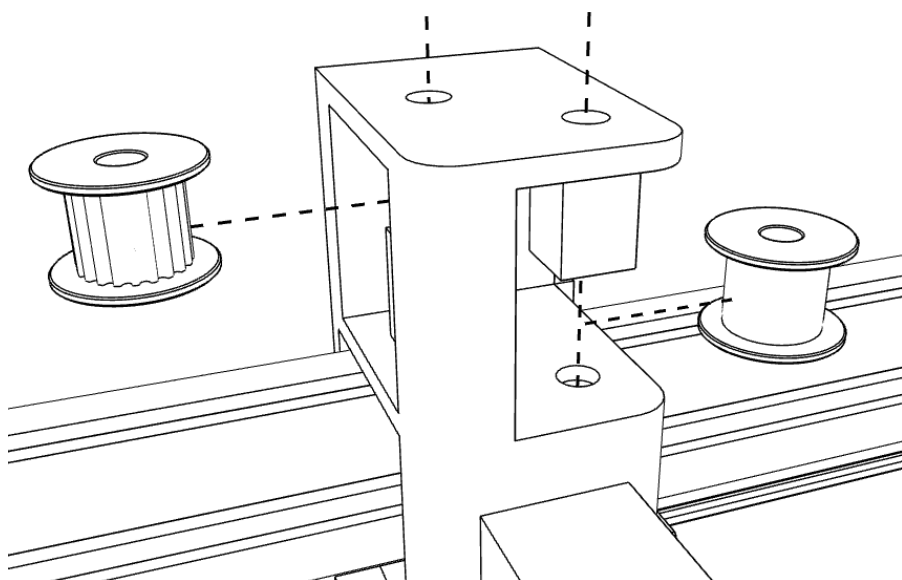
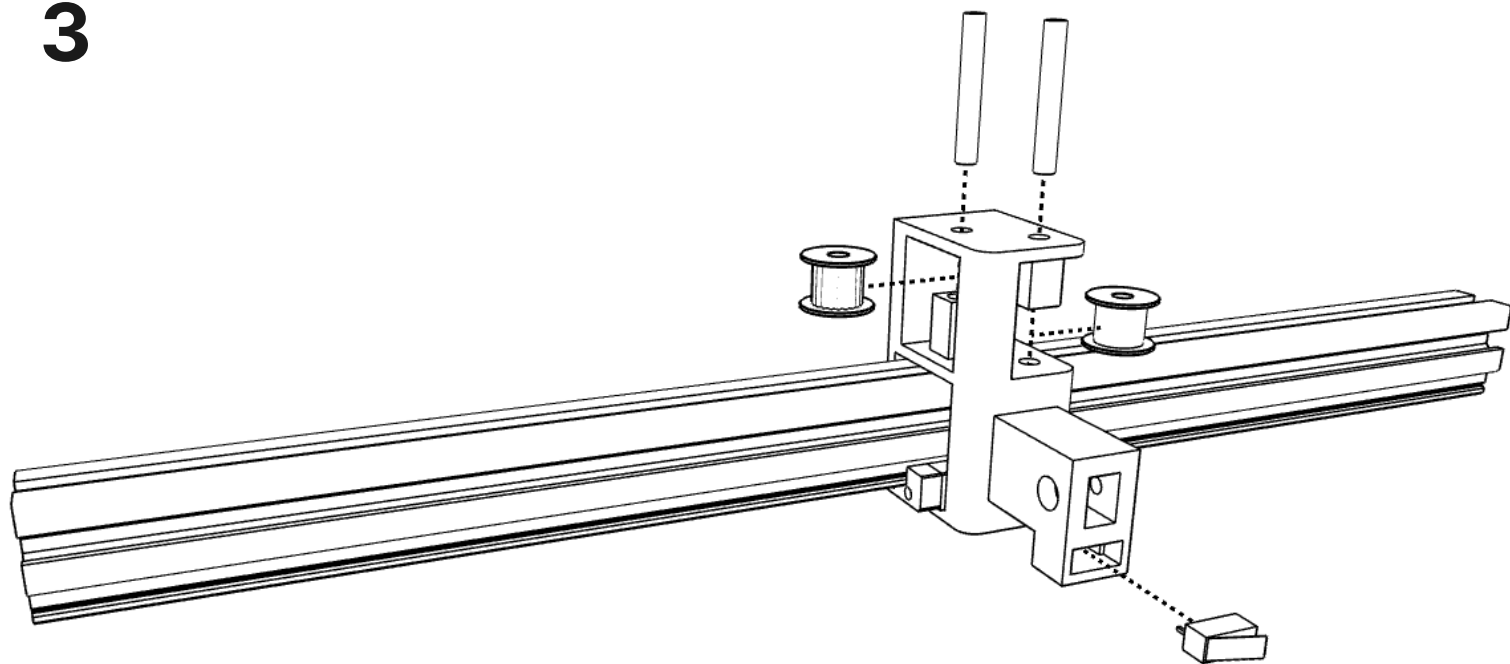
1



2



3

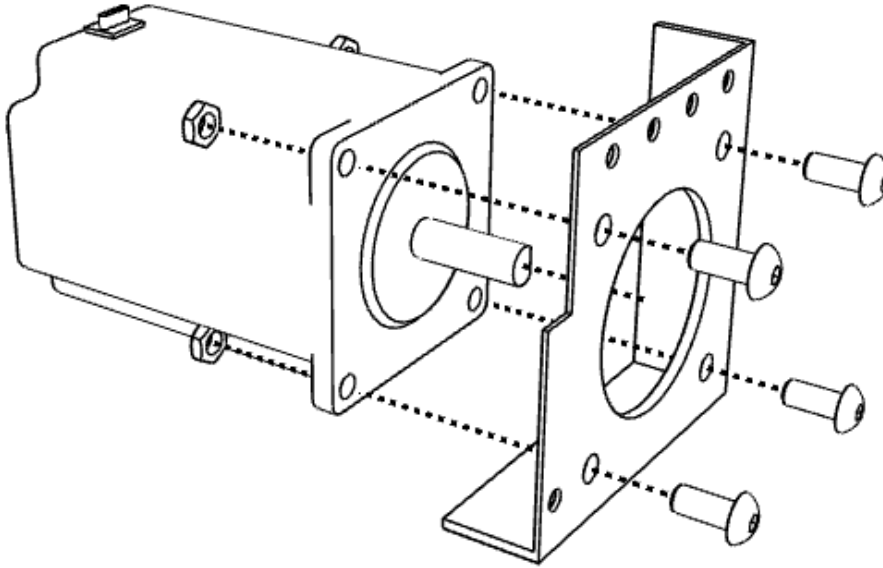


6. CoreXY Drive Module Left Side

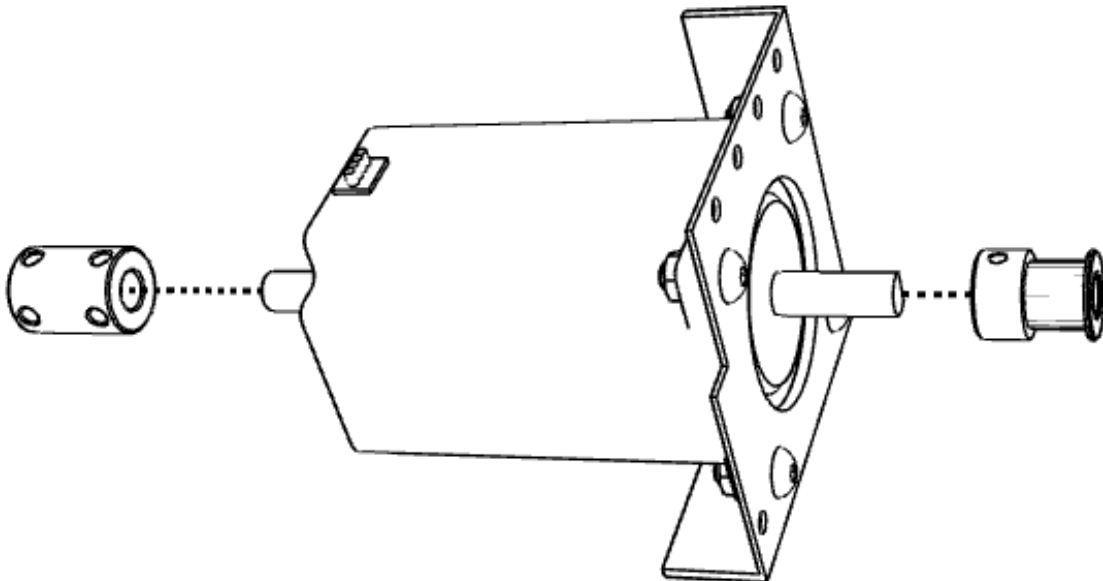
Required Components:

- CoreXY Motor Mount Left x1
- Motor Plate Left x1
- NEMA 23 Stepper Motor x1
- Rotary Encoder x1
- GT2 Drive Pulley x1
- 6×6 mm Flexible Coupling x1
- M3×10 mm Screw x7
- M3 Hex Nut x5
- M5×12 mm Screw x4
- M5 Hex Nut x4

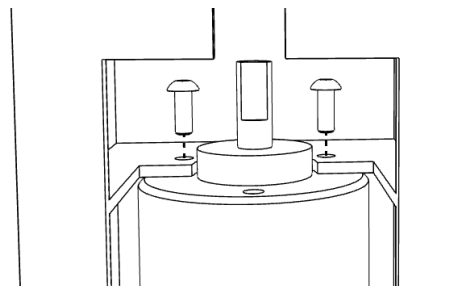
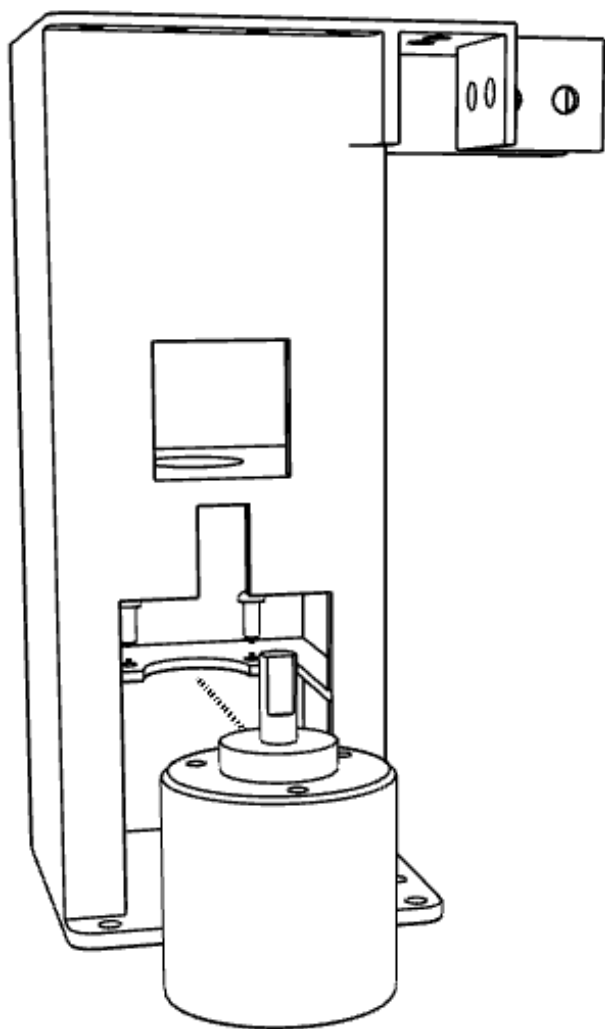
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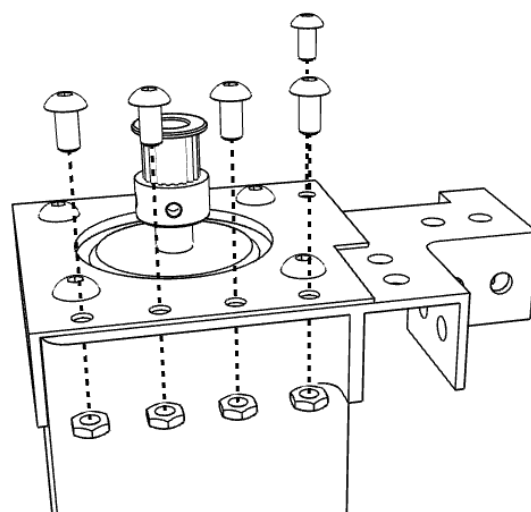
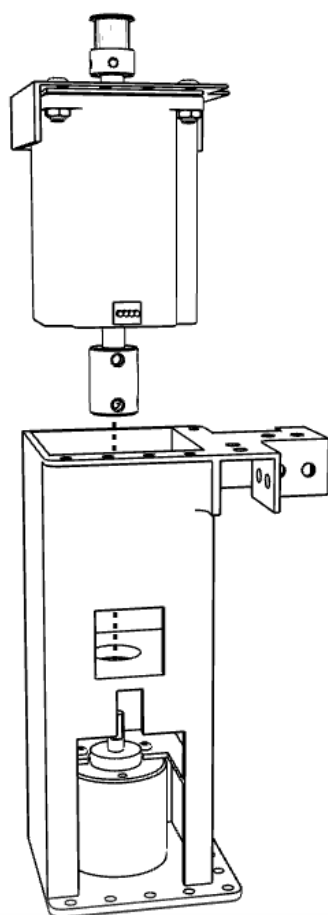
2



3



4

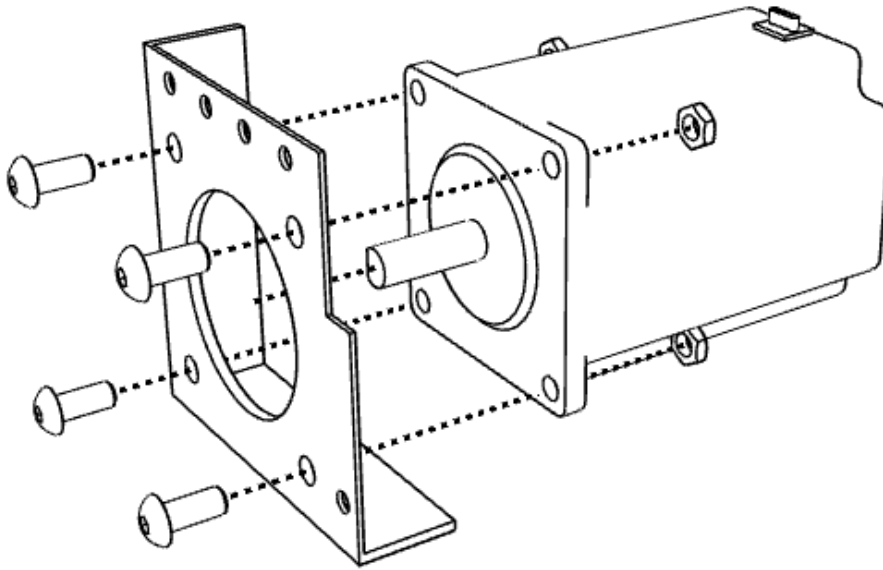


7. CoreXY Drive Module Right Side

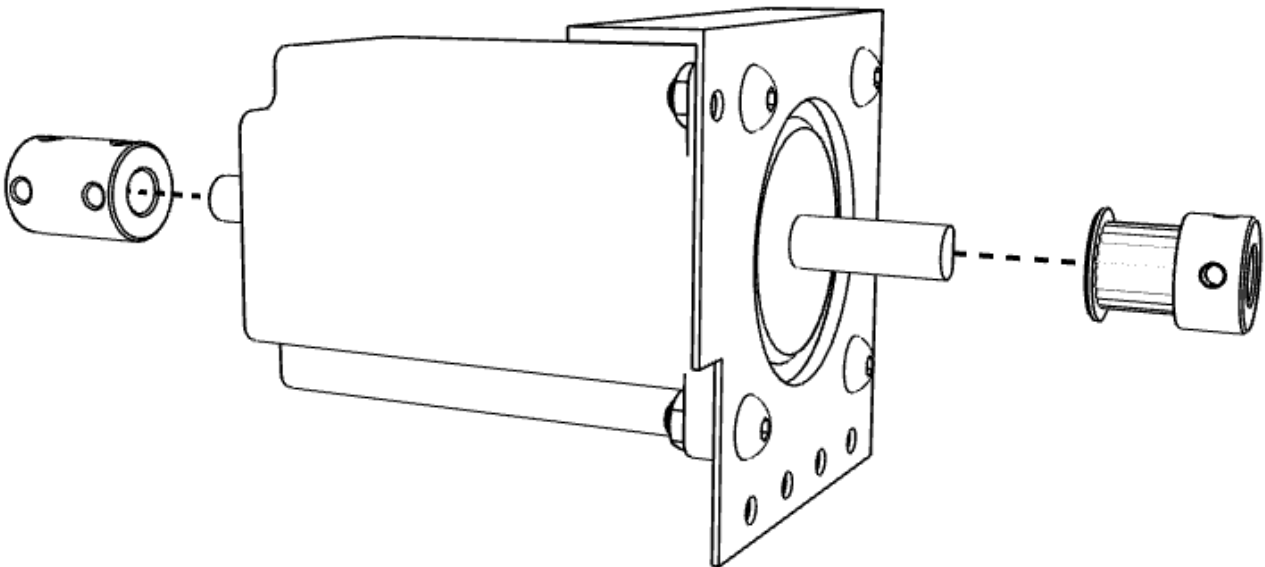
Required Components:

- CoreXY Motor Mount Right x1
- Motor Plate Right x1
- NEMA 23 Stepper Motor x1
- Rotary Encoder x1
- GT2 Drive Pulley x1
- Limit Switch x1
- 6×6 mm Flexible Coupling x1
- M3×10 mm Screw x7
- M3 Hex Nut x5
- M5×12 mm Screw x4
- M5 Hex Nut x4

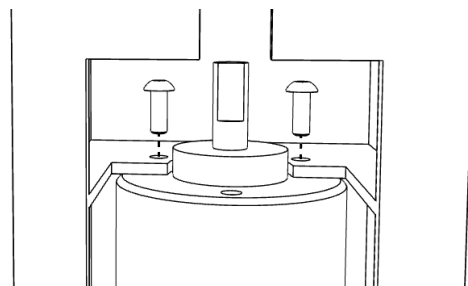
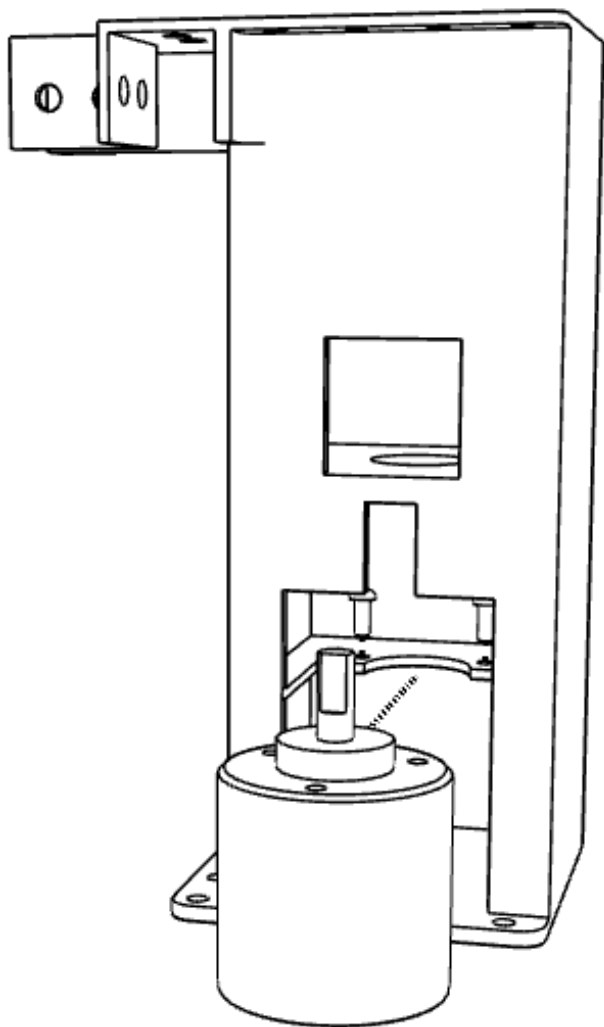
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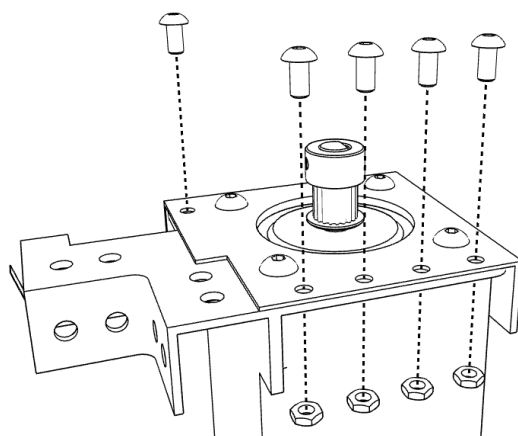
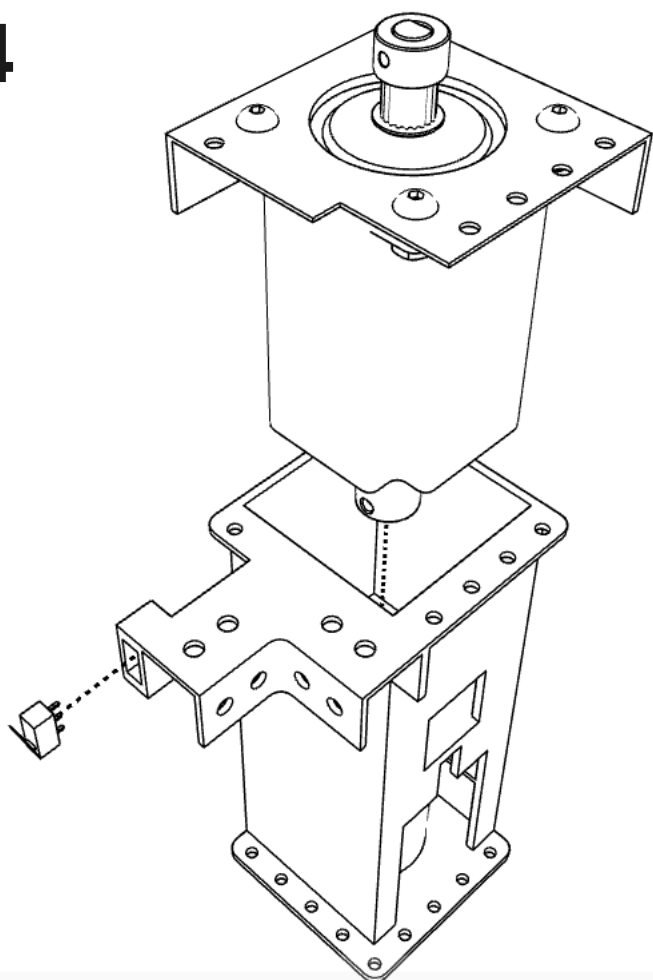
2



3



4

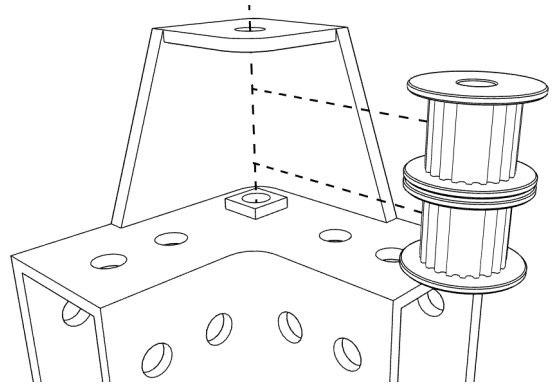
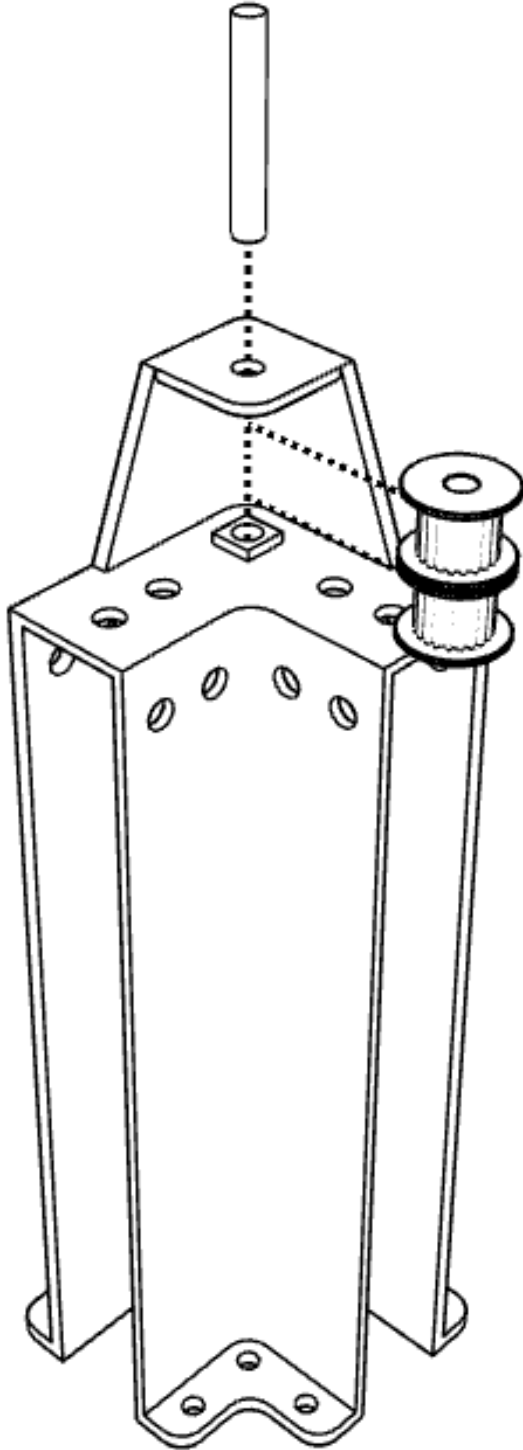


8. Idler Pulley Module

Required Components:

- Idler Pulley Housing x2
- GT2 Toothed Idler Pulley x4
- Ø5×10 mm Linear Shaft x2

1



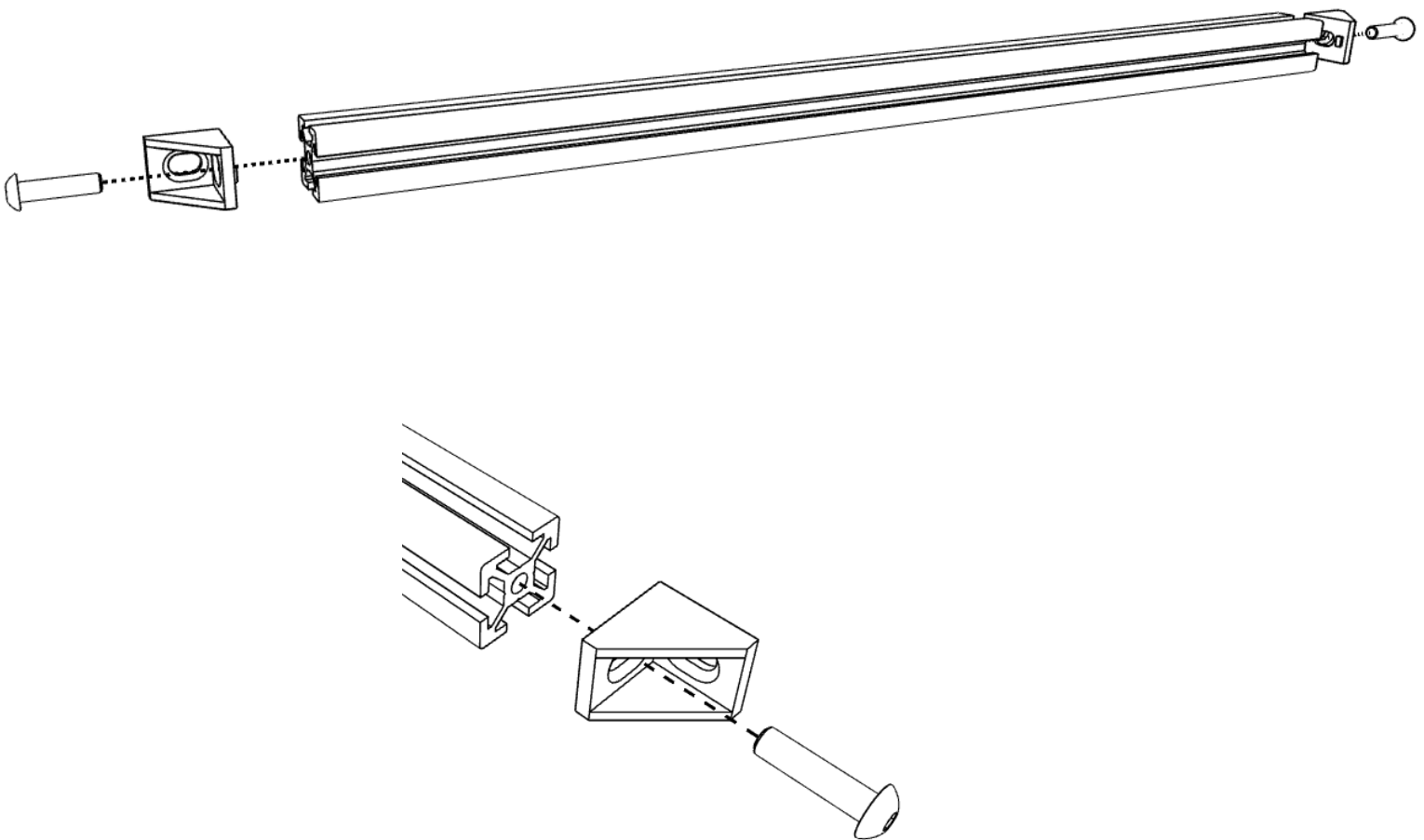
2x

9. Frame Profile Assembly

Required Components:

- 400 mm Aluminum Structural Profile x2
- T-Slot Corner Bracket x4
- M5×10 mm Screw x4

1



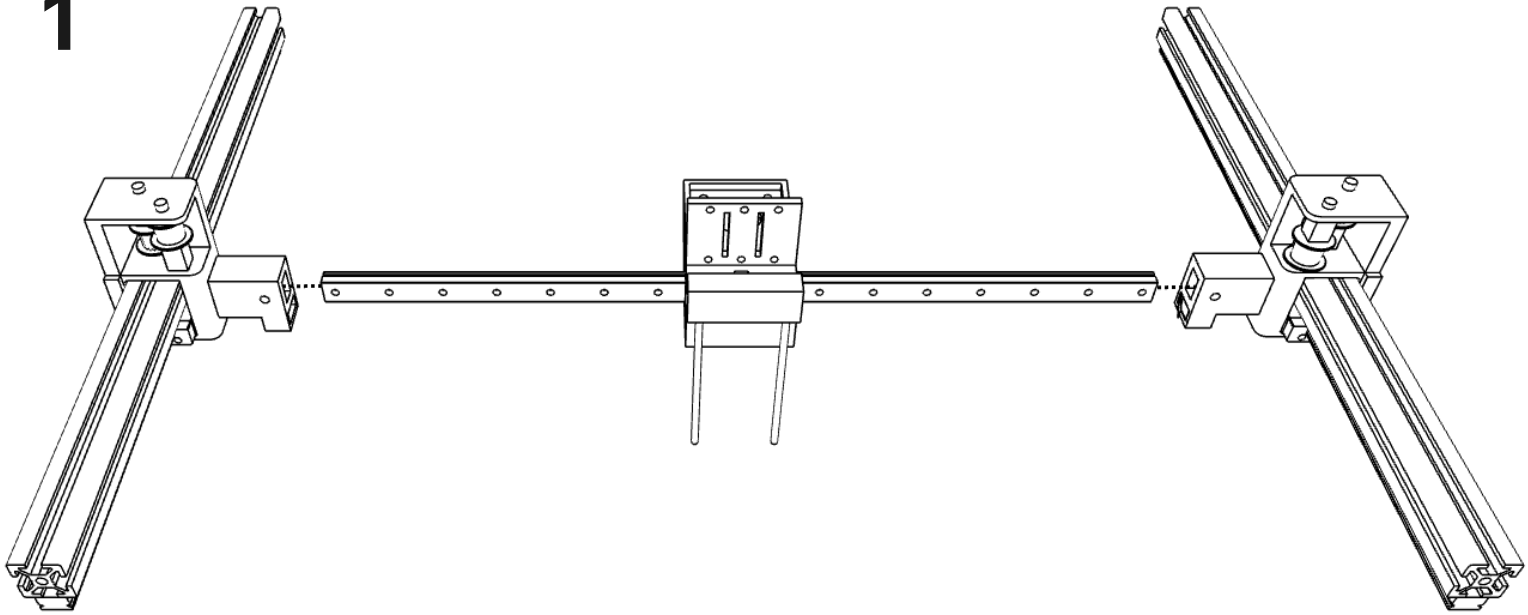
2x

10. Main Frame Integration

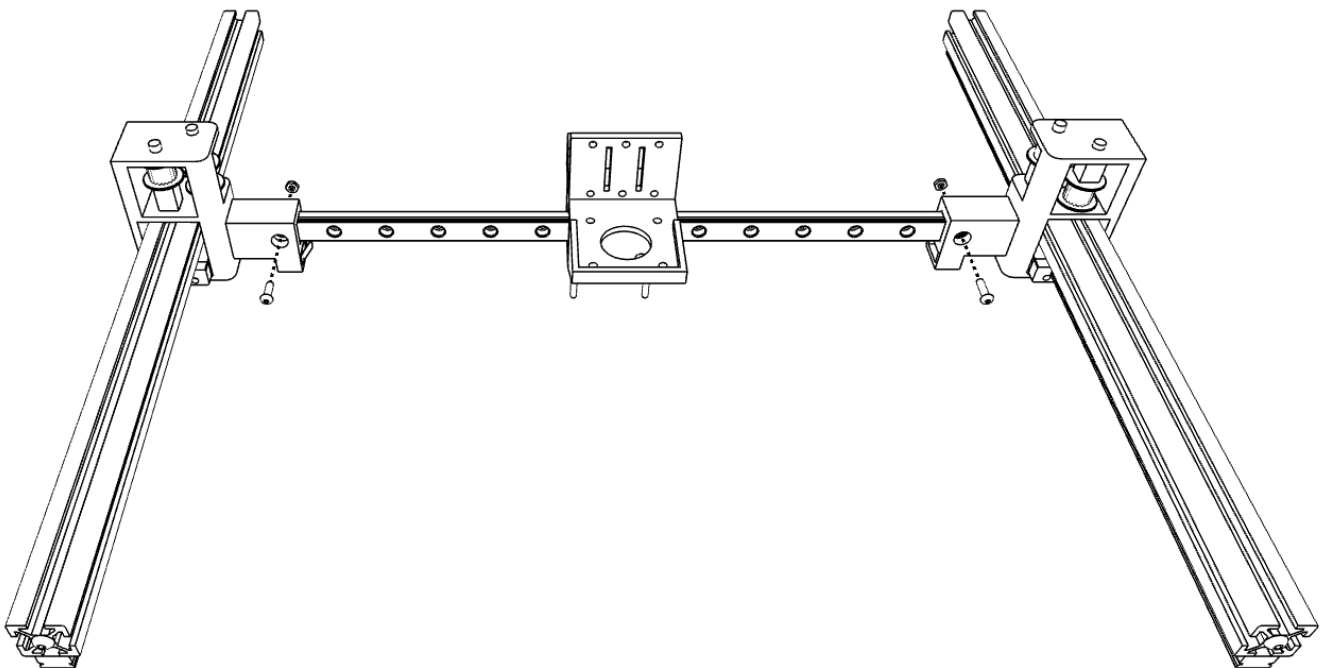
Required Components:

- Assembled Module from Step 4
 - Assembled Module from Step 5
 - Assembled Module from Step 6
 - Assembled Module from Step 7
 - Assembled Module from Step 8
 - Assembled Module from Step 9
 - M3×10 mm Screw x2
 - M3 Hex Nut x2
 - M5×10 mm Screw x44
 - M5 T-Nut x42
-

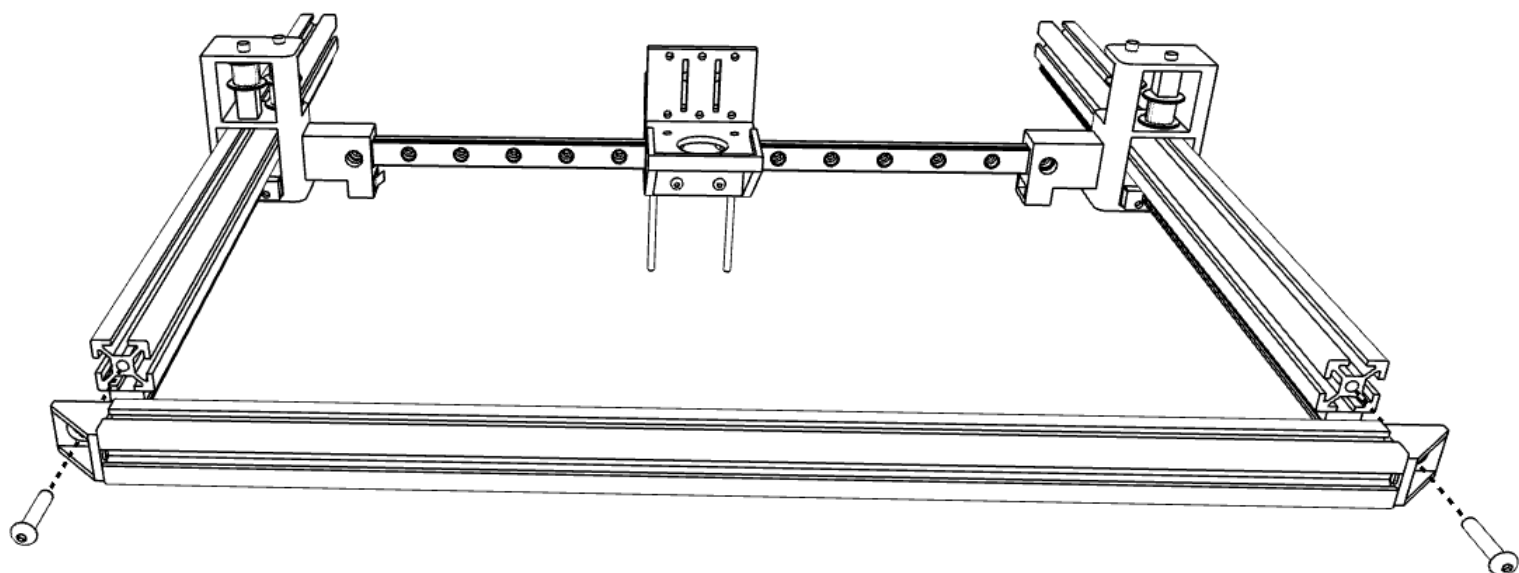
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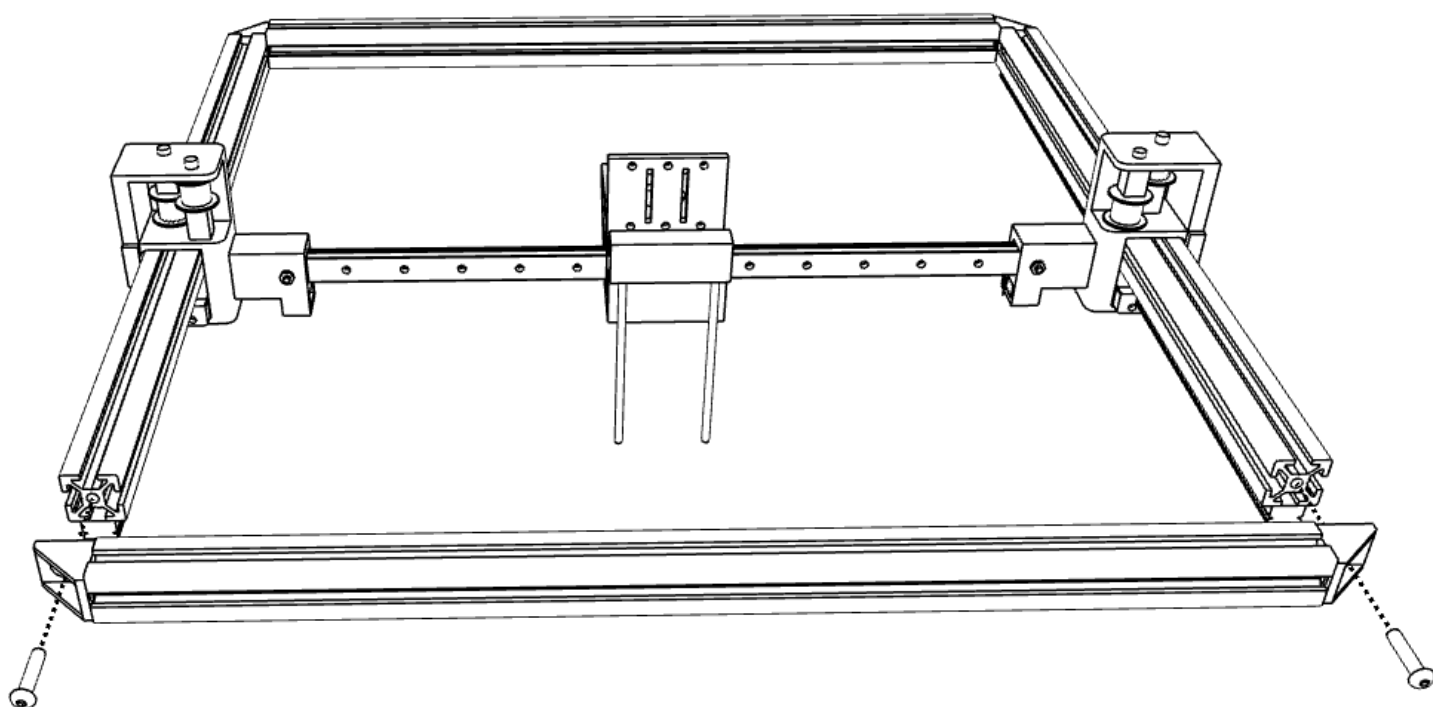
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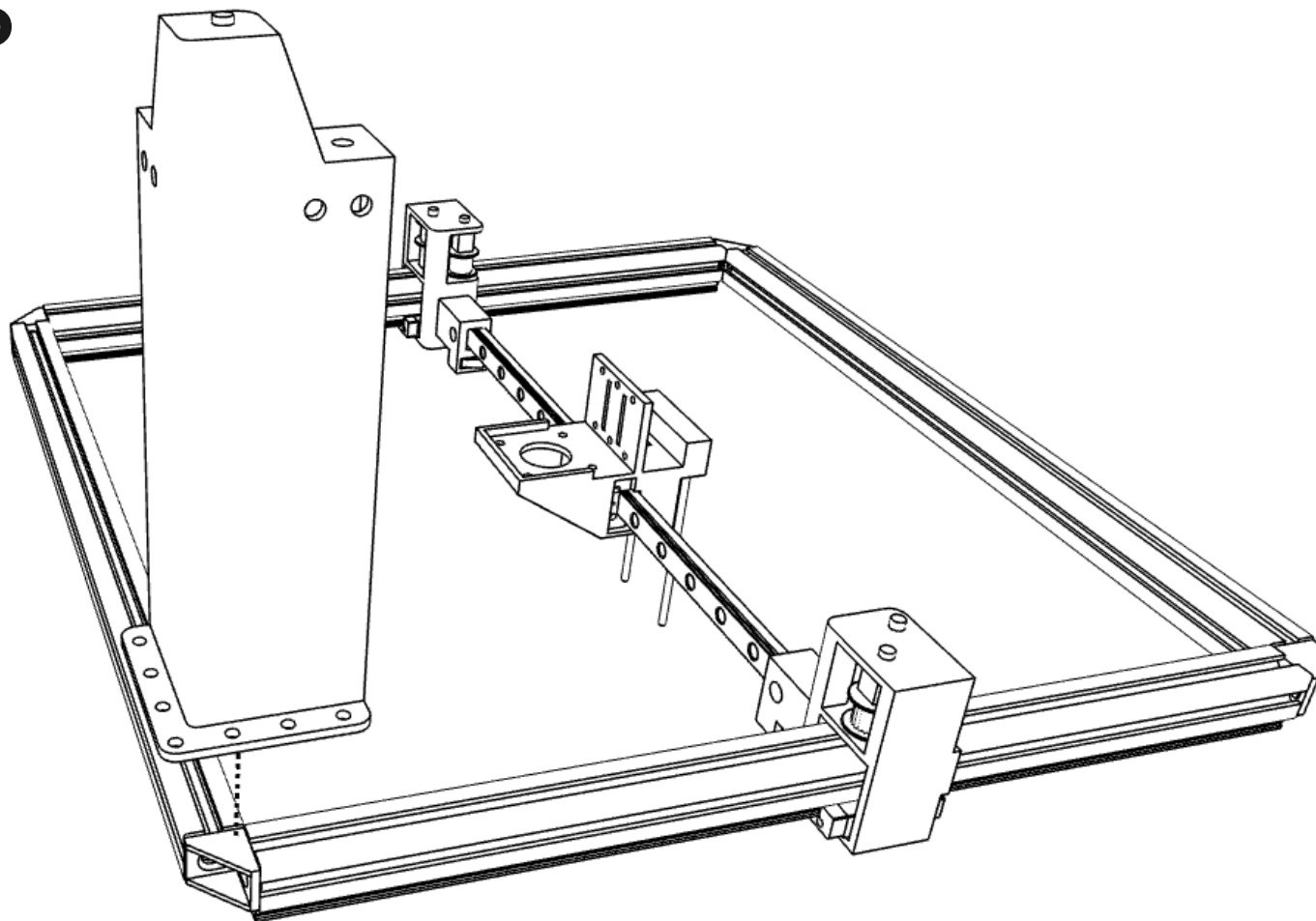
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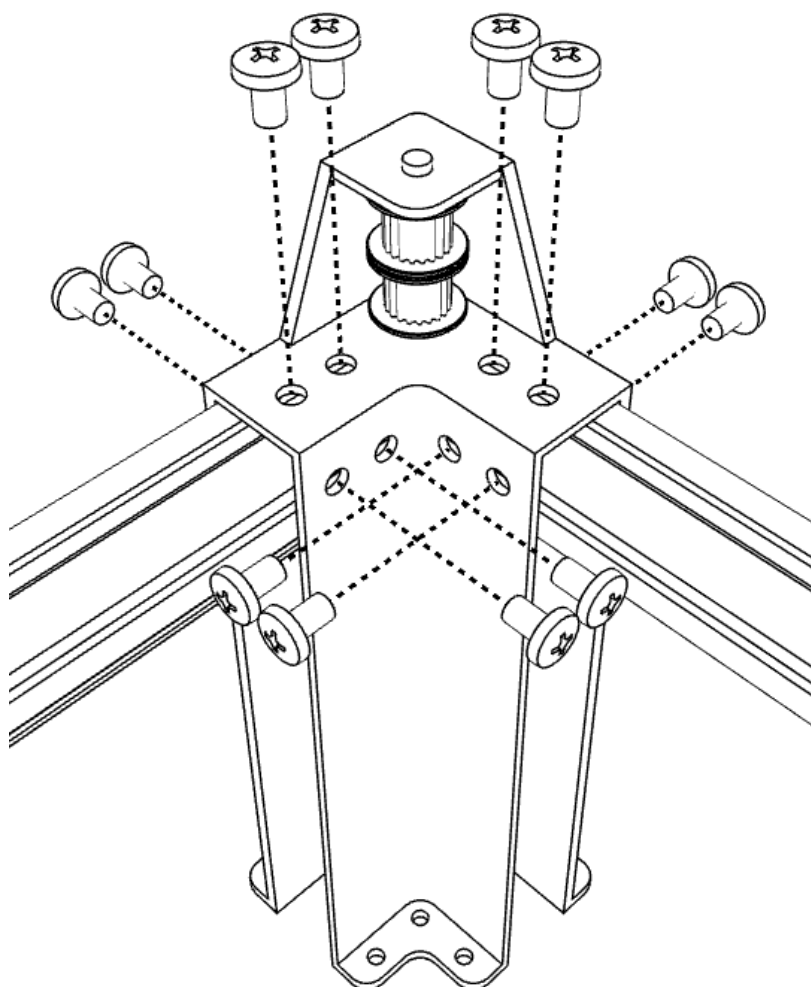
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5

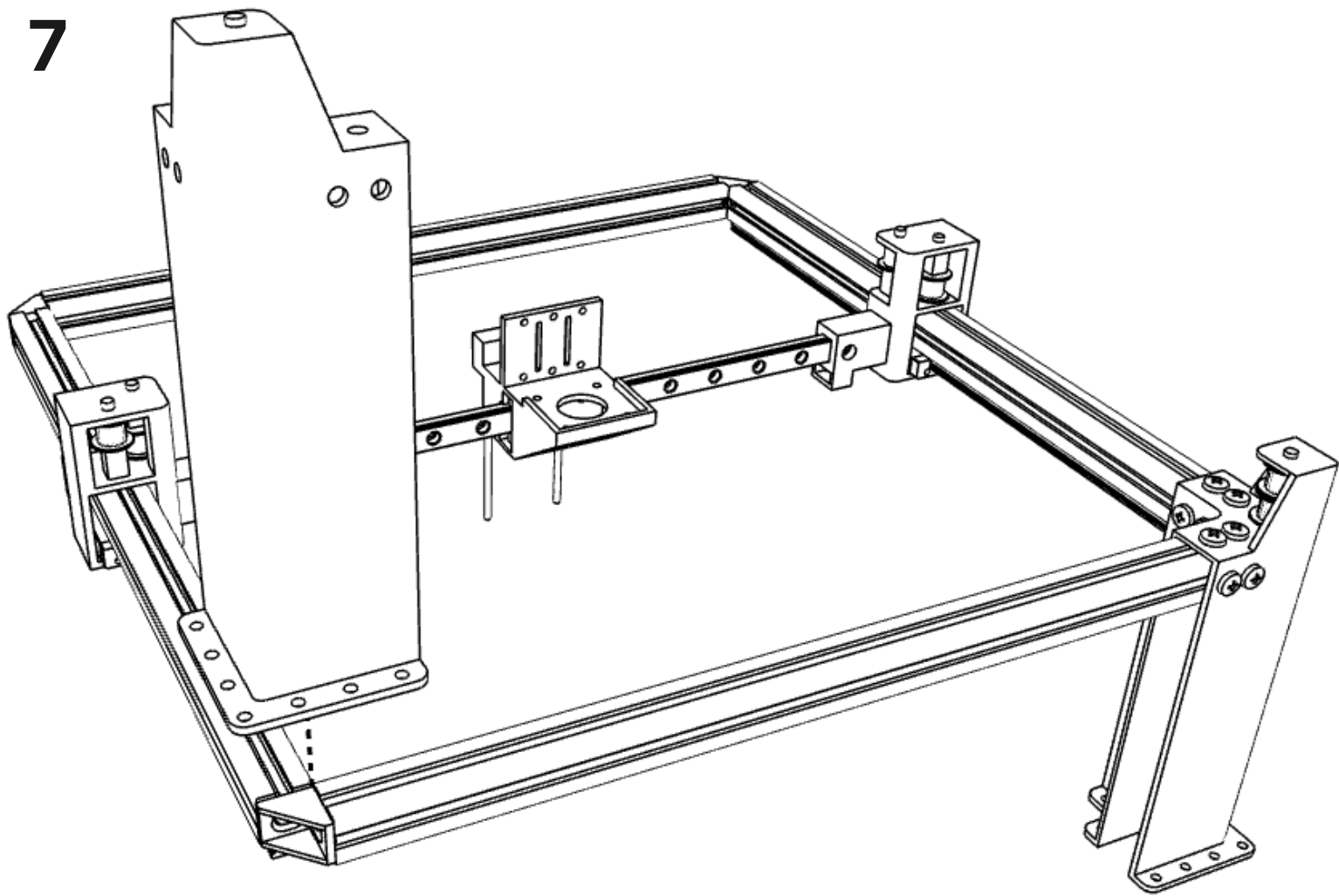


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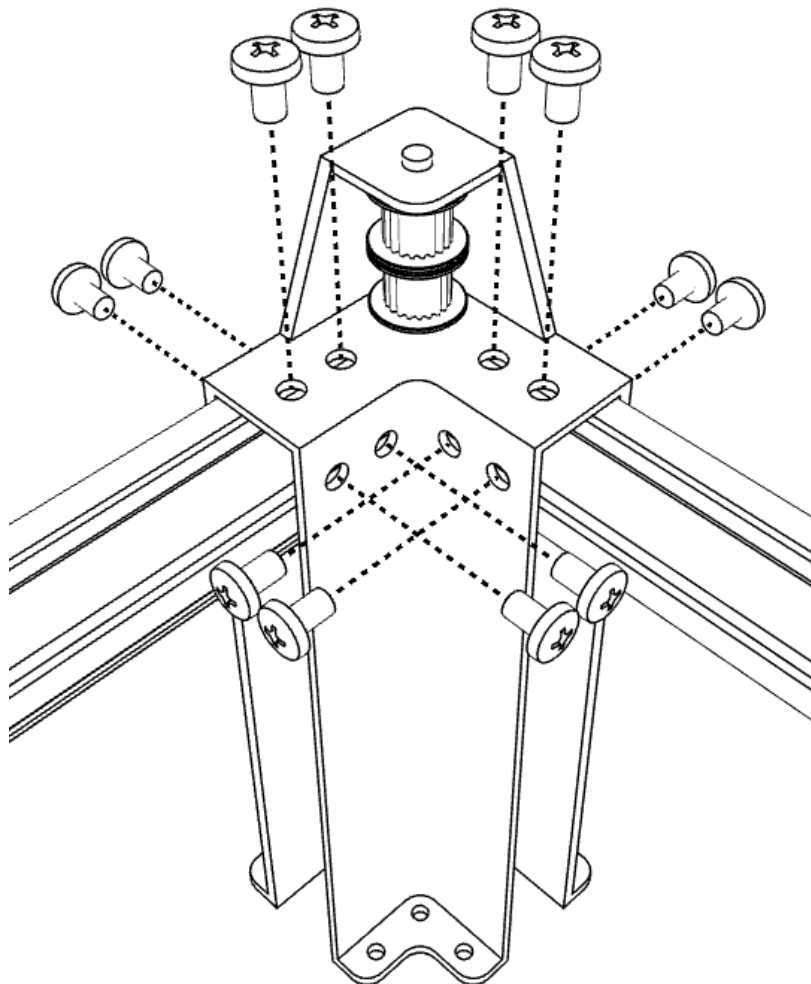


For each screw, insert
an M5 T-nut into the
profile.

7

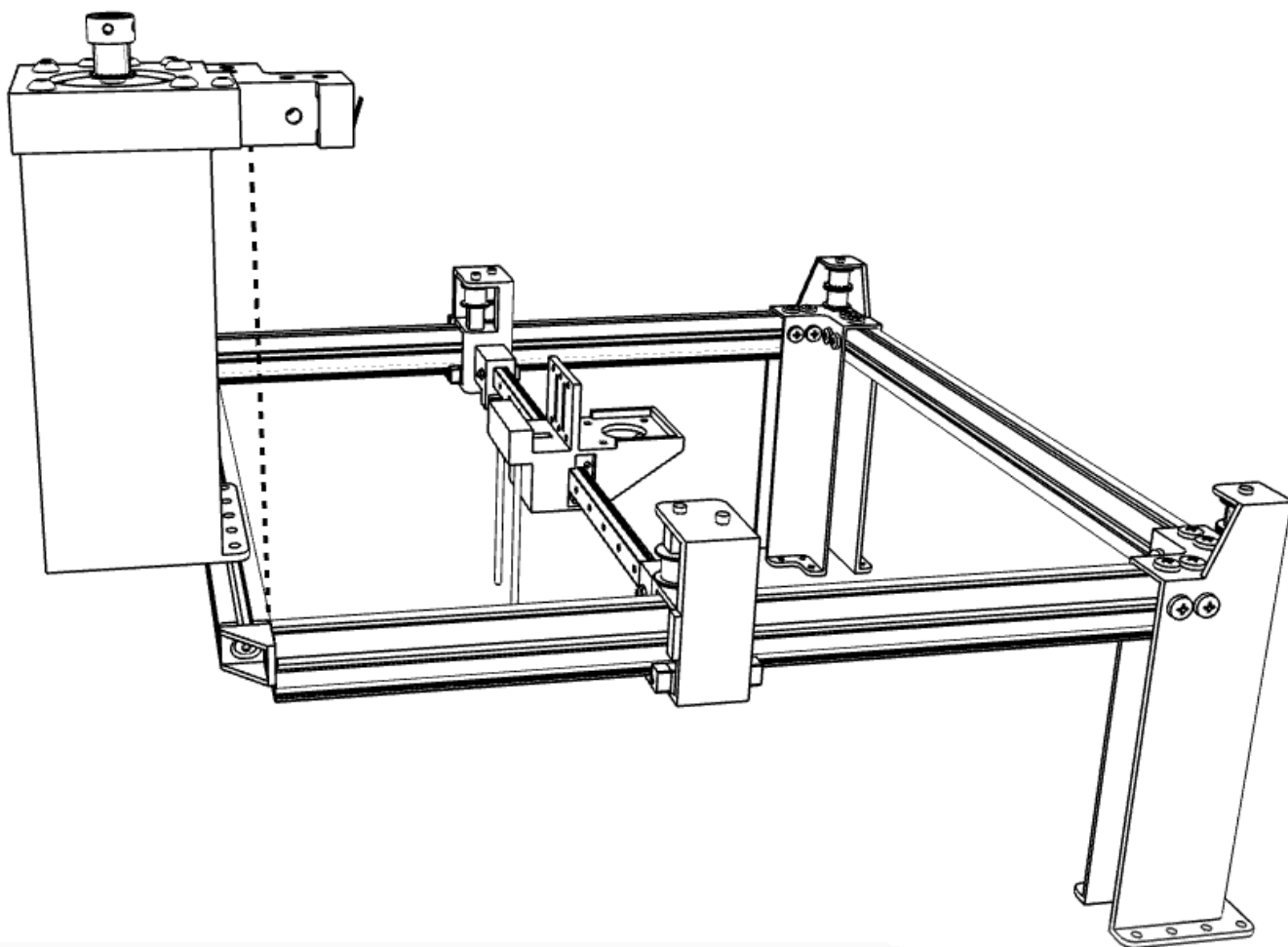


8

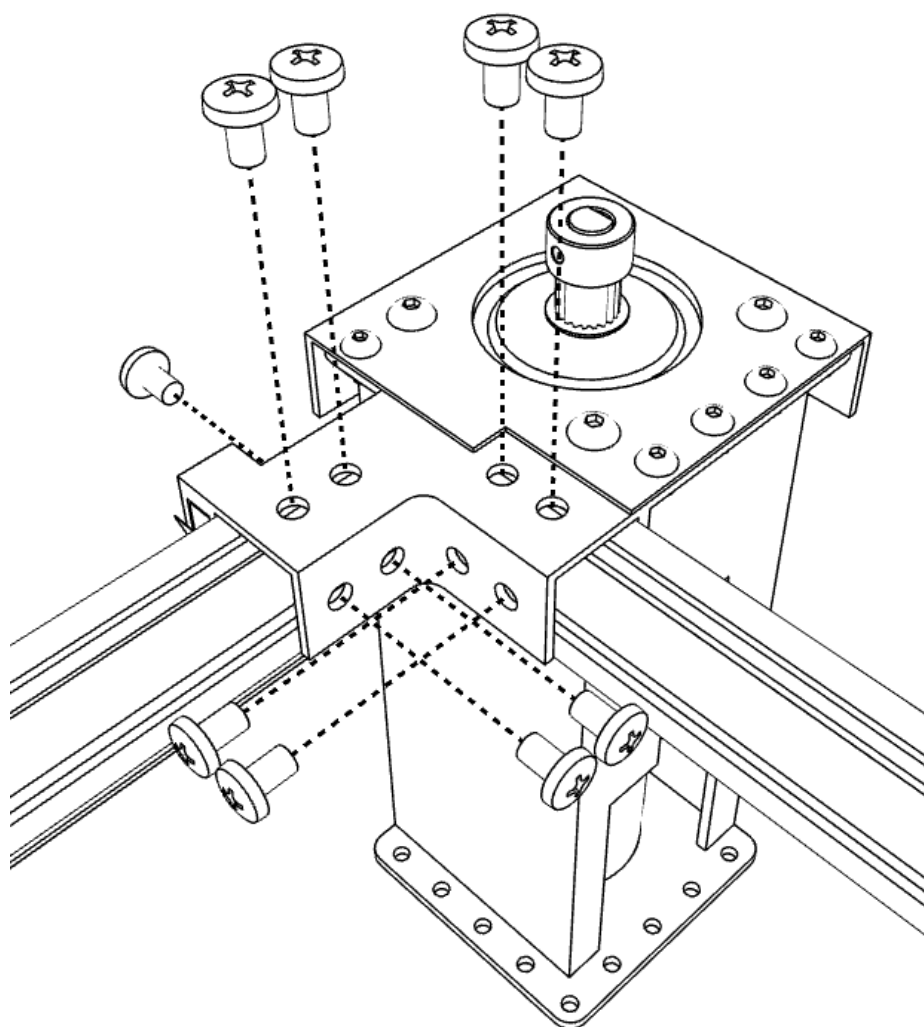


For each screw, insert
an M5 T-nut into the
profile.

9

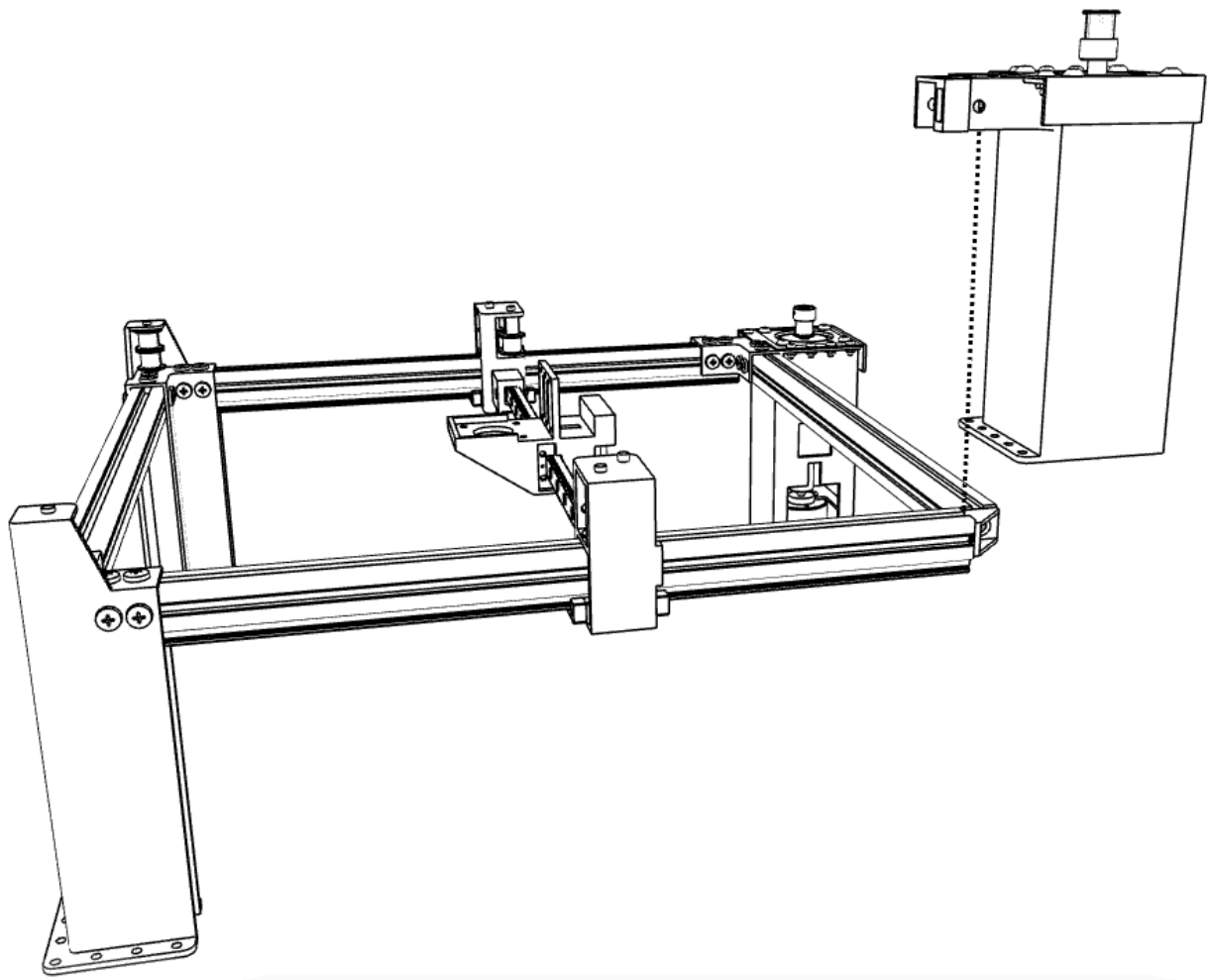


10

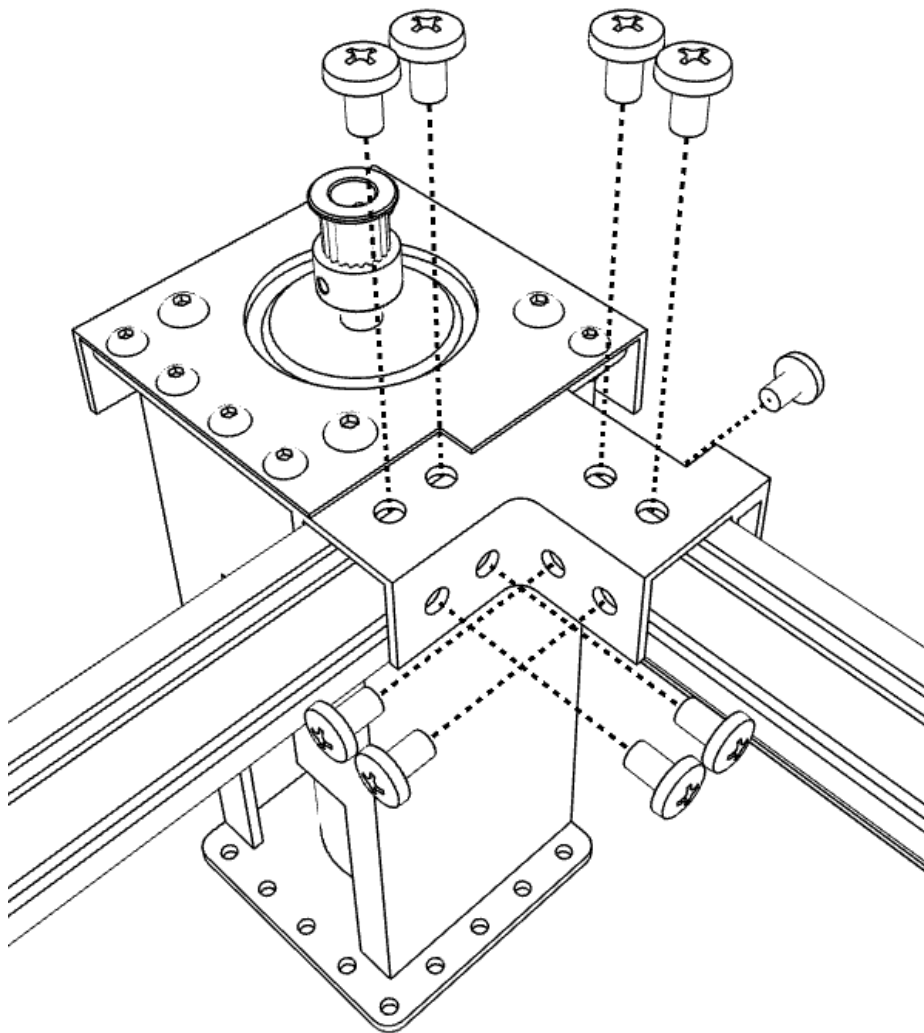


For each screw, insert an M5 T-nut into the profile.

11



12



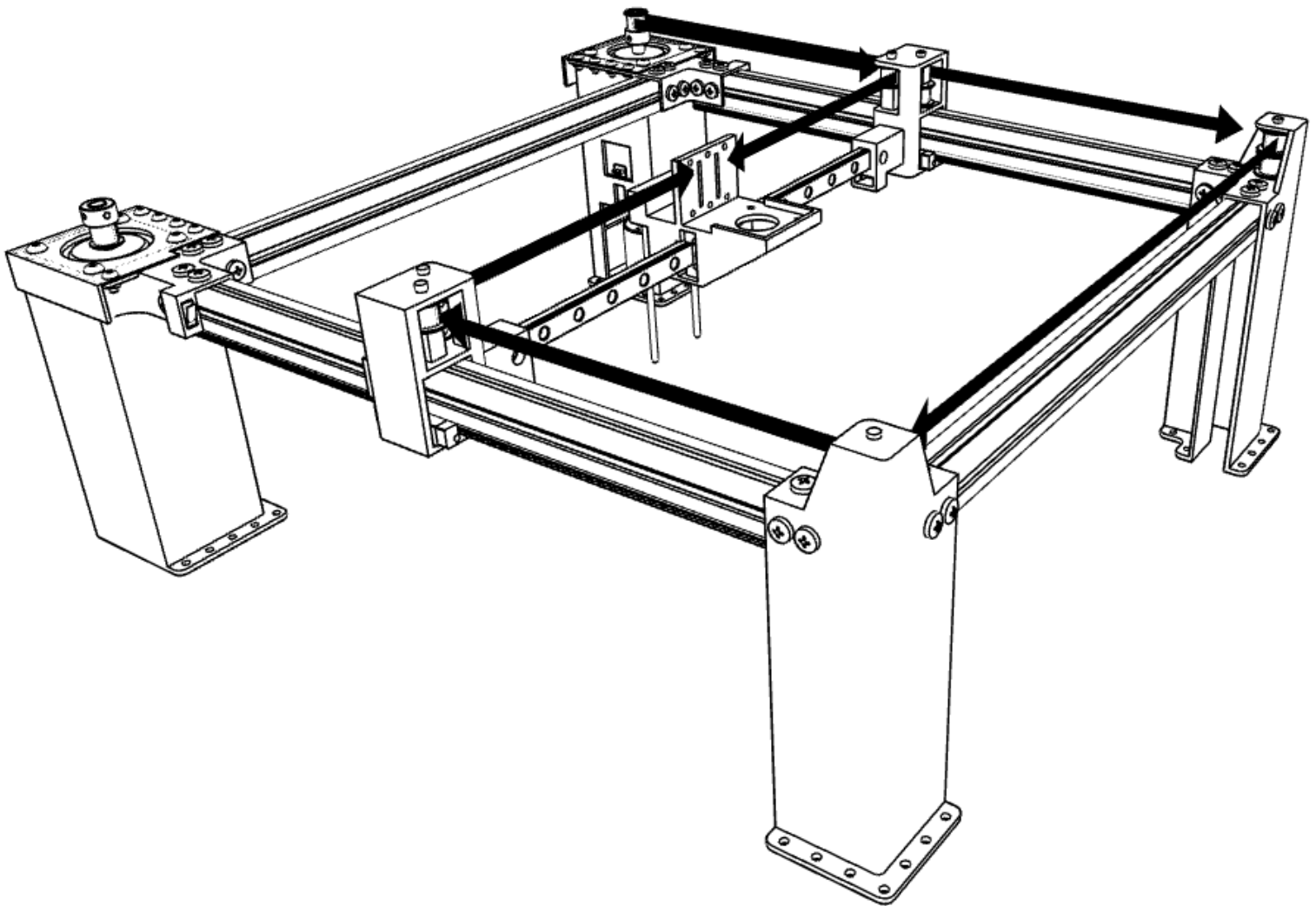
For each screw, insert an M5 T-nut into the profile.

11. Timing Belt Routing and Tensioning

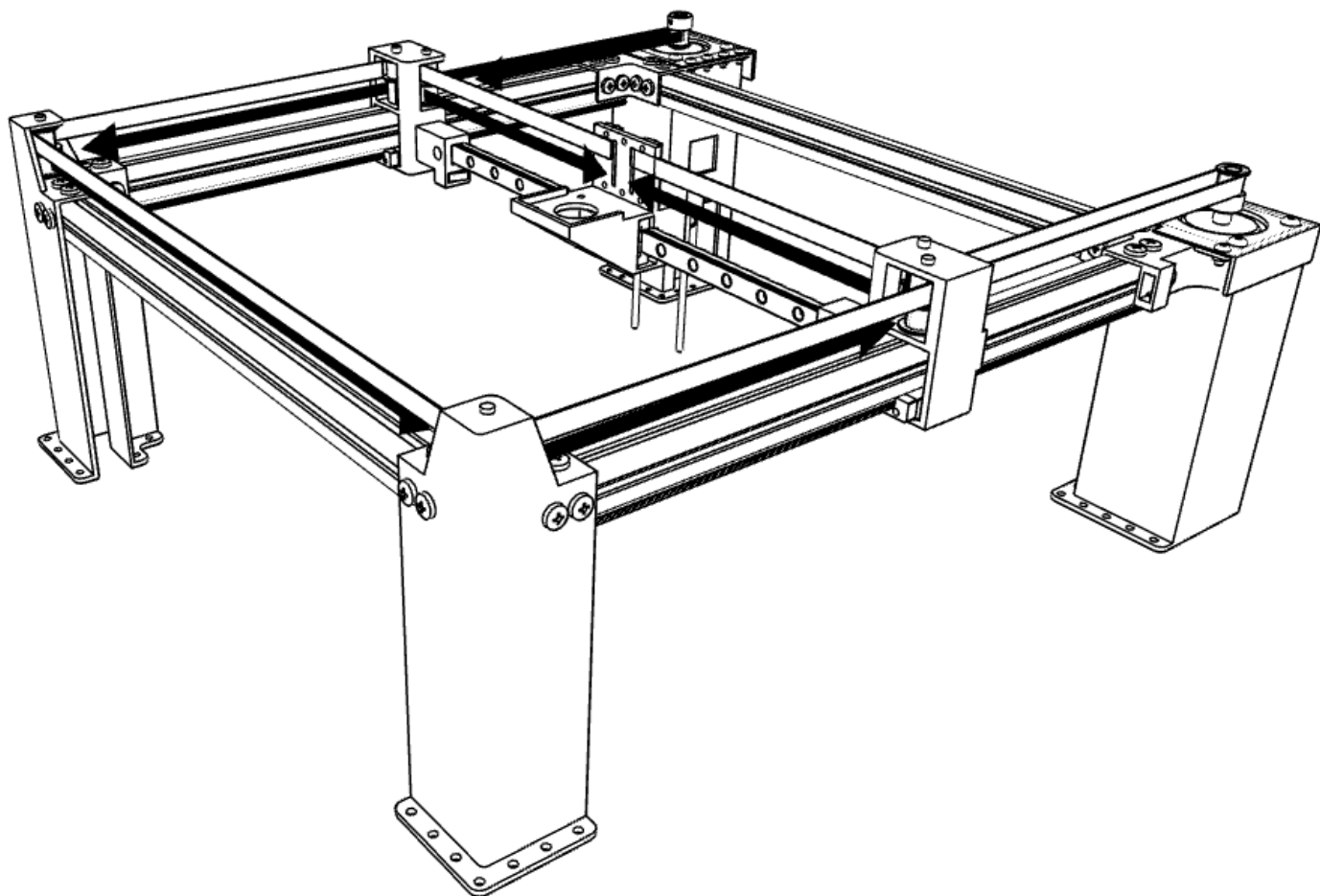
Required Components:

- Module from Step 10
- GT2 2000 mm Timing Belt x2
- Timing Belt Clamp Plate x4
- M3×10 mm Screw x4
- M3 Hex Nut x4

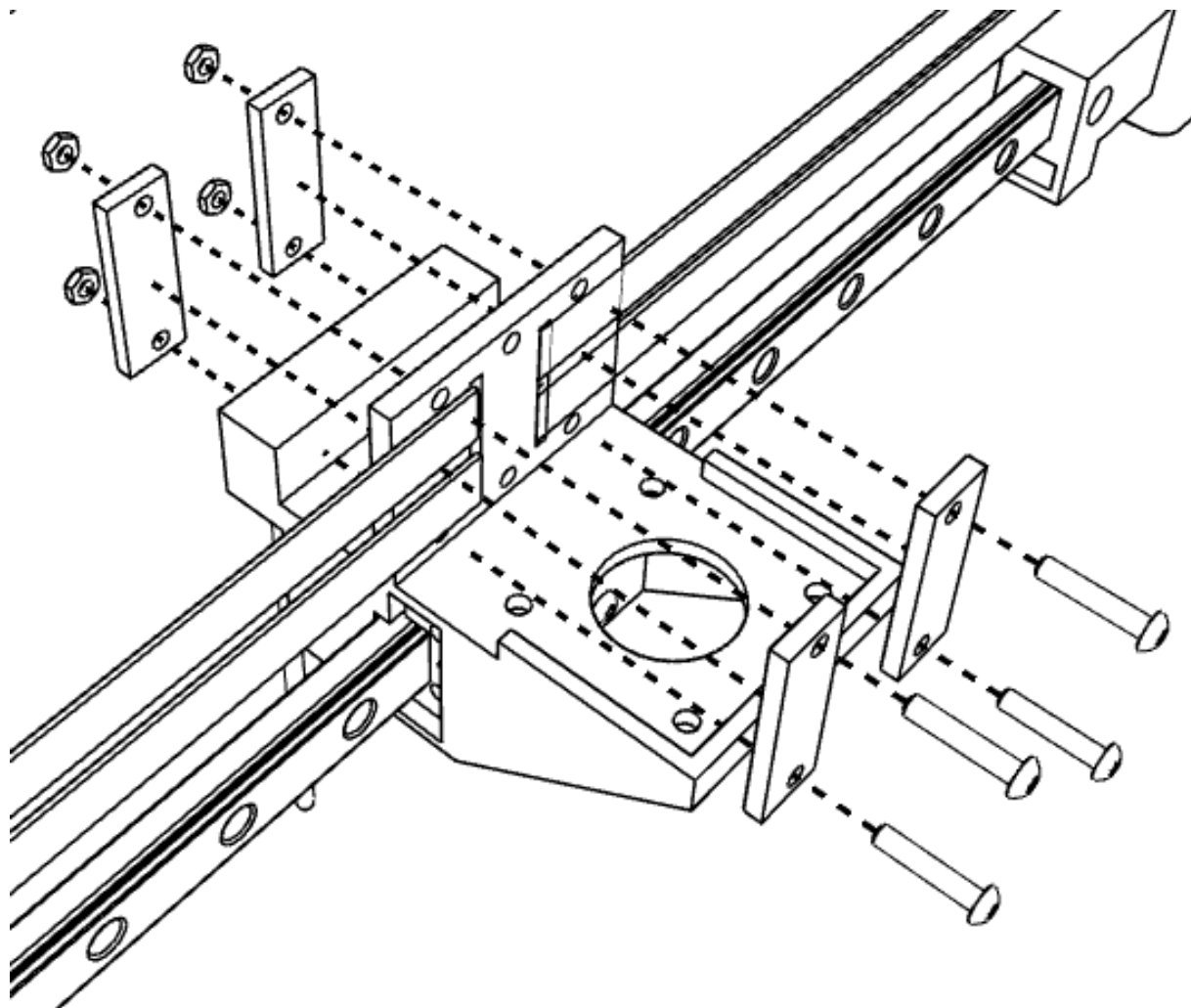
1



2



3

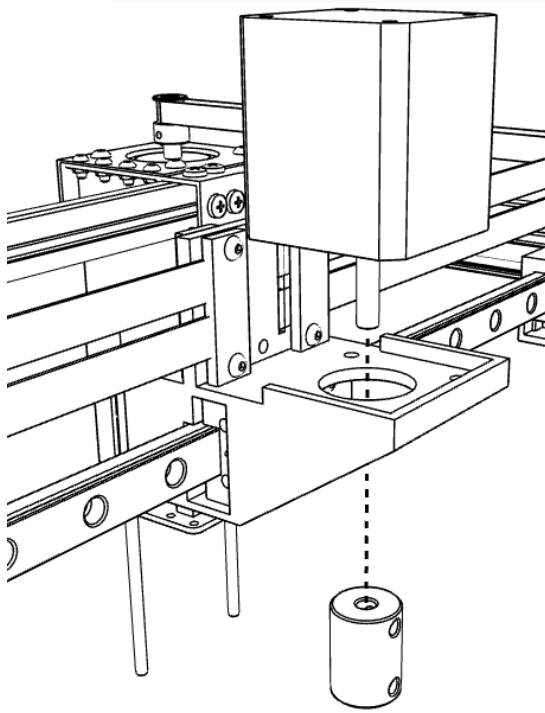


12. Z-Axis Drive Assembly

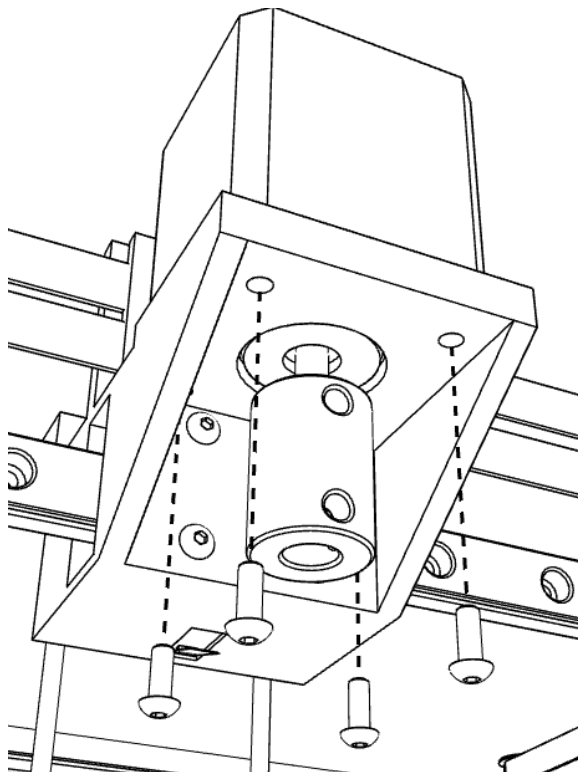
Required Components:

- Module from Step 11
- NEMA 17 Stepper Motor x1
- 5×8 mm Rigid Coupling x1
- M3 × 6 mm Screw x4

1



2

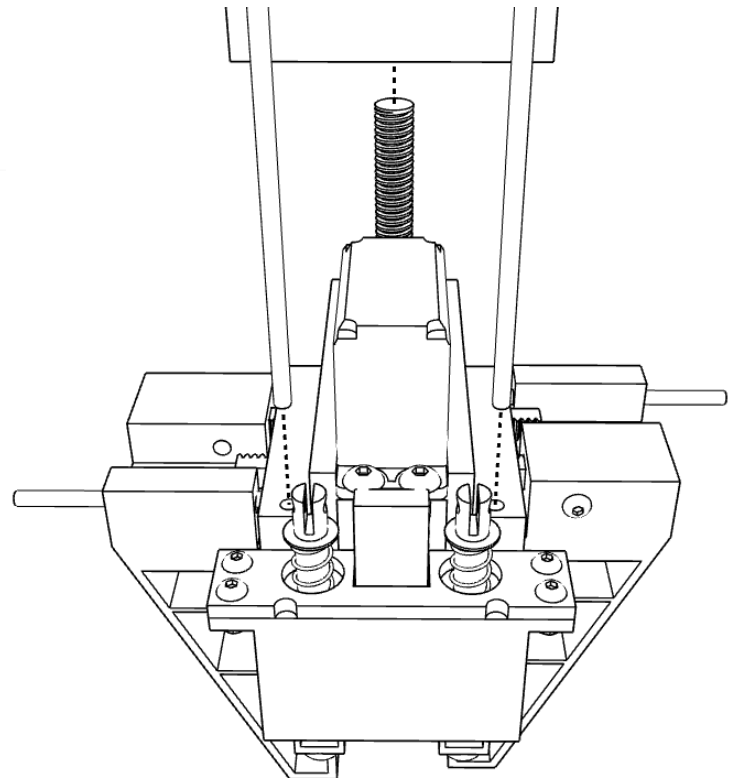
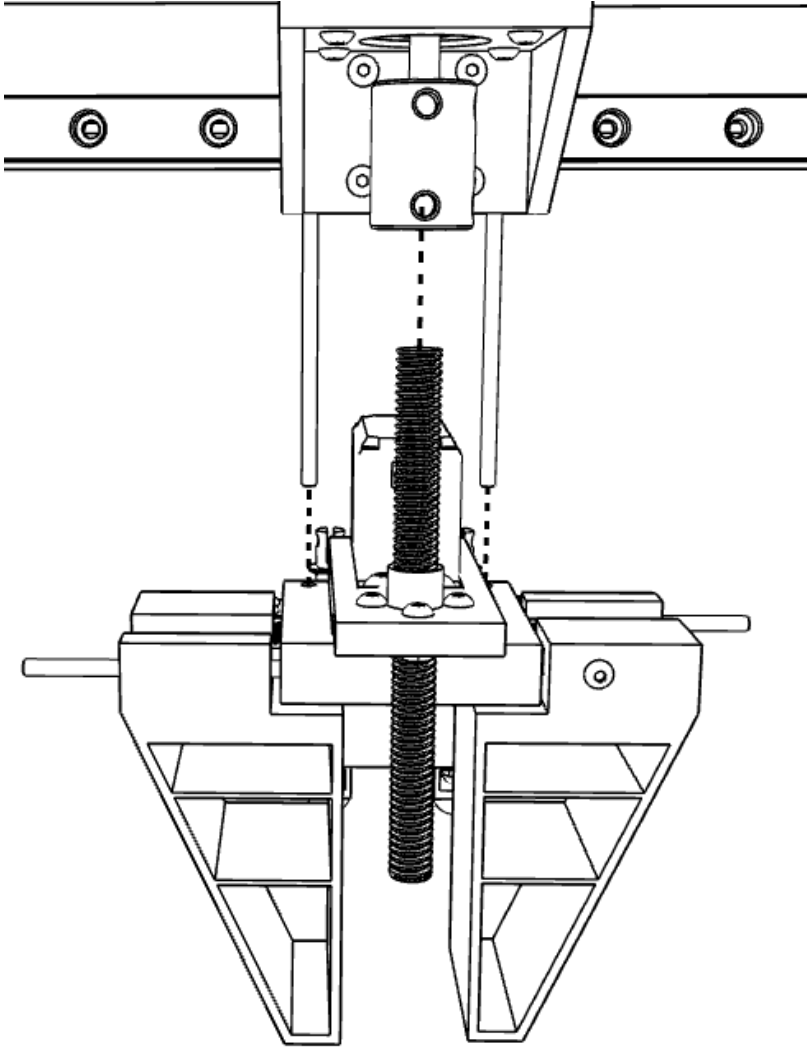


13. Effector and Platform Integration

Required Components:

- Module from Step 1
- Module from Step 12

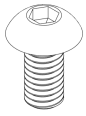
1



14. List of Elements:

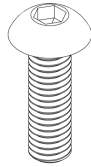
Fasteners:

M3×6 mm Screw



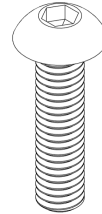
4x

M3×10 mm Screw



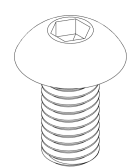
76x

M3×16 mm Screw



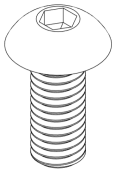
2x

M5×10 mm Screw



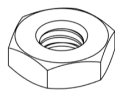
48x

M5×12 mm Screw



8x

M3 Hex Nut



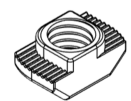
28x

M5 Hex Nut



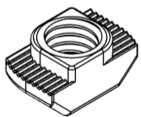
8x

M3 T-Nut



32x

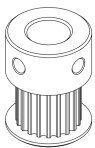
M5 T-Nut



42x

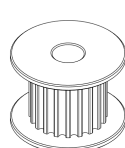
Drive Components:

GT2 Drive Pulley



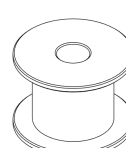
2x

GT2 Toothed Idler Pulley



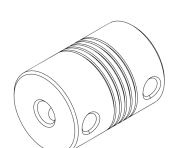
6x

GT2 Smooth Idler Pulley



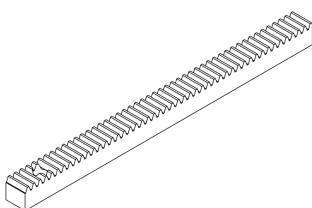
2x

6×6 mm Flexible Coupling



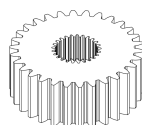
2x

Gear Rack



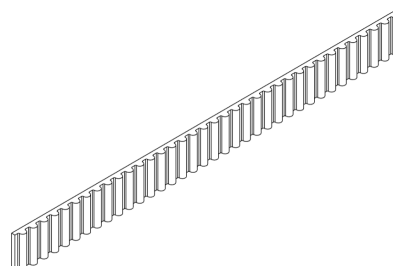
2x

Servo Gear



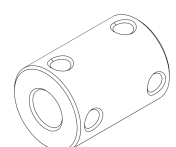
1x

GT2 2000 mm Timing Belt



2x

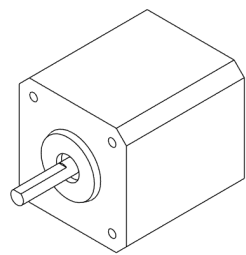
5×8 mm Rigid Coupling



1x

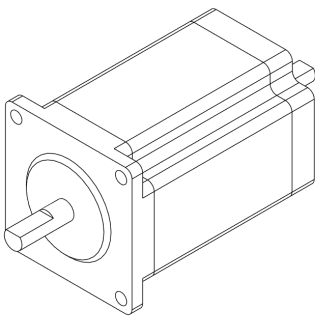
Electronics:

NEMA 17 Stepper Motor



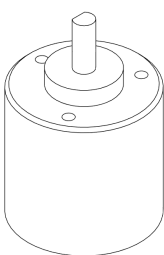
1x

NEMA 23 Stepper Motor



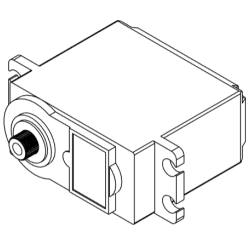
2x

Rotary Encoder



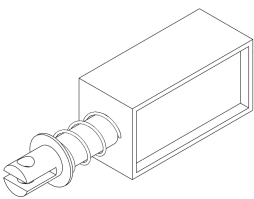
2x

Servo Motor



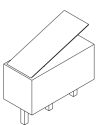
1x

Solenoid



2x

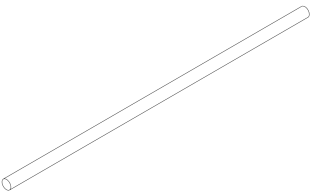
Limit Switch



3x

Structural Parts:

Ø3×130 mm Linear Shaft



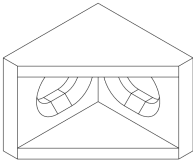
4x

Ø5×10 mm Linear Shaft



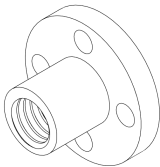
6x

T-Slot Corner Bracket



4x

TR8×2 Lead Screw Nut



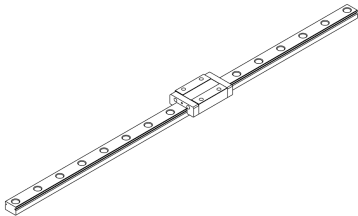
1x

Solenoid Tip



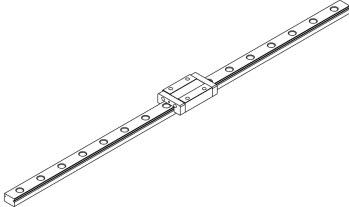
2x

MGN12 386 mm
Linear Guide Rail



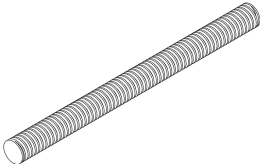
1x

MGN12 400 mm
Linear Guide Rail



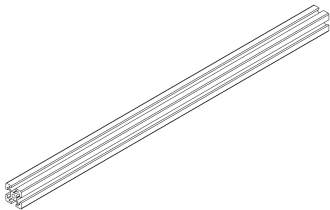
2x

TR8×2 100 mm Lead Screw



1x

400 mm Aluminum Profile



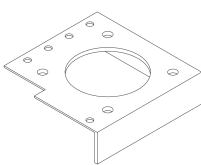
4x

Timing Belt Clamp Plate



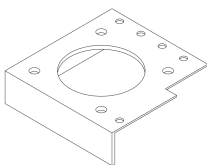
4x

Motor Plate Left



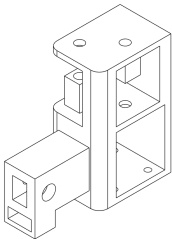
1x

Motor Plate Right



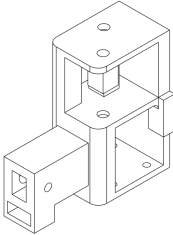
1x

Linear Carriage Body Left



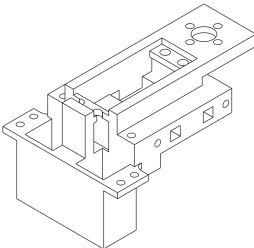
1x

Linear Carriage Body Right



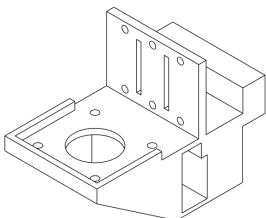
1x

Jaw Mount Body



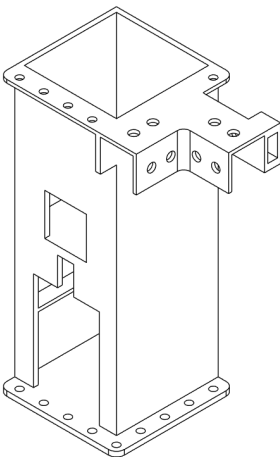
1x

Gripper Carriage Body



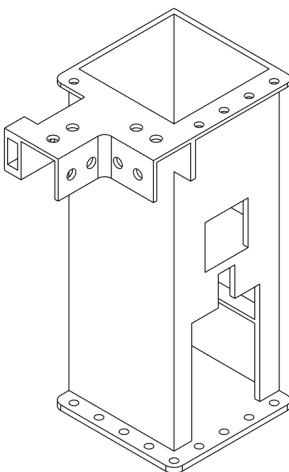
1x

CoreXY Motor Mount Left



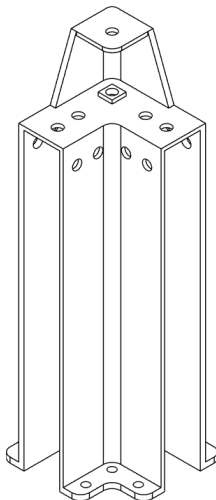
1x

CoreXY Motor Mount Right



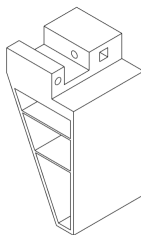
1x

Idler Pulley Housing



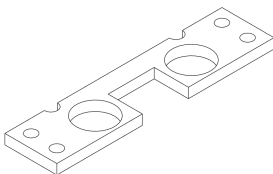
2x

Gripper Jaws



2x

Solenoid Cover



1x